

WorkshopPlus: Azure AI Platform and Services

Azure AI Fundamentals

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Learning Units covered in this Module



Azure AI Services Overview



Azure AI Speech



Azure AI Language



Azure AI Vision / Azure AI Content Understanding



Azure AI Search



Azure OpenAI / Azure AI Content Safety



Azure AI Fundamentals

Objectives

After completing this Learning, you will be able to:

Understand....

What the Azure AI Platform is.

The AI Services available in the Azure AI Platform.

How to leverage the Azure AI Services to built applications.

The capabilities of each service.



Azure AI Services Overview

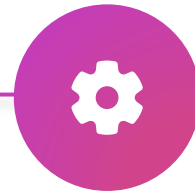
AI-driven business transformation



**Enrich
employee
experiences**



**Reinvent
customer
engagement**



**Reshape
business
processes**



**Bend the
curve on
innovation**

Azure AI Foundry



Copilot Studio



Visual Studio



GitHub



Azure AI
Foundry SDK



Model Catalog

Foundational models

Open-source models

Task models

Industry models



Azure
OpenAI Service



Azure AI
Search



Azure AI
Agent Service



Azure AI
Content Safety



Azure
Machine Learning

Evaluations

Customization

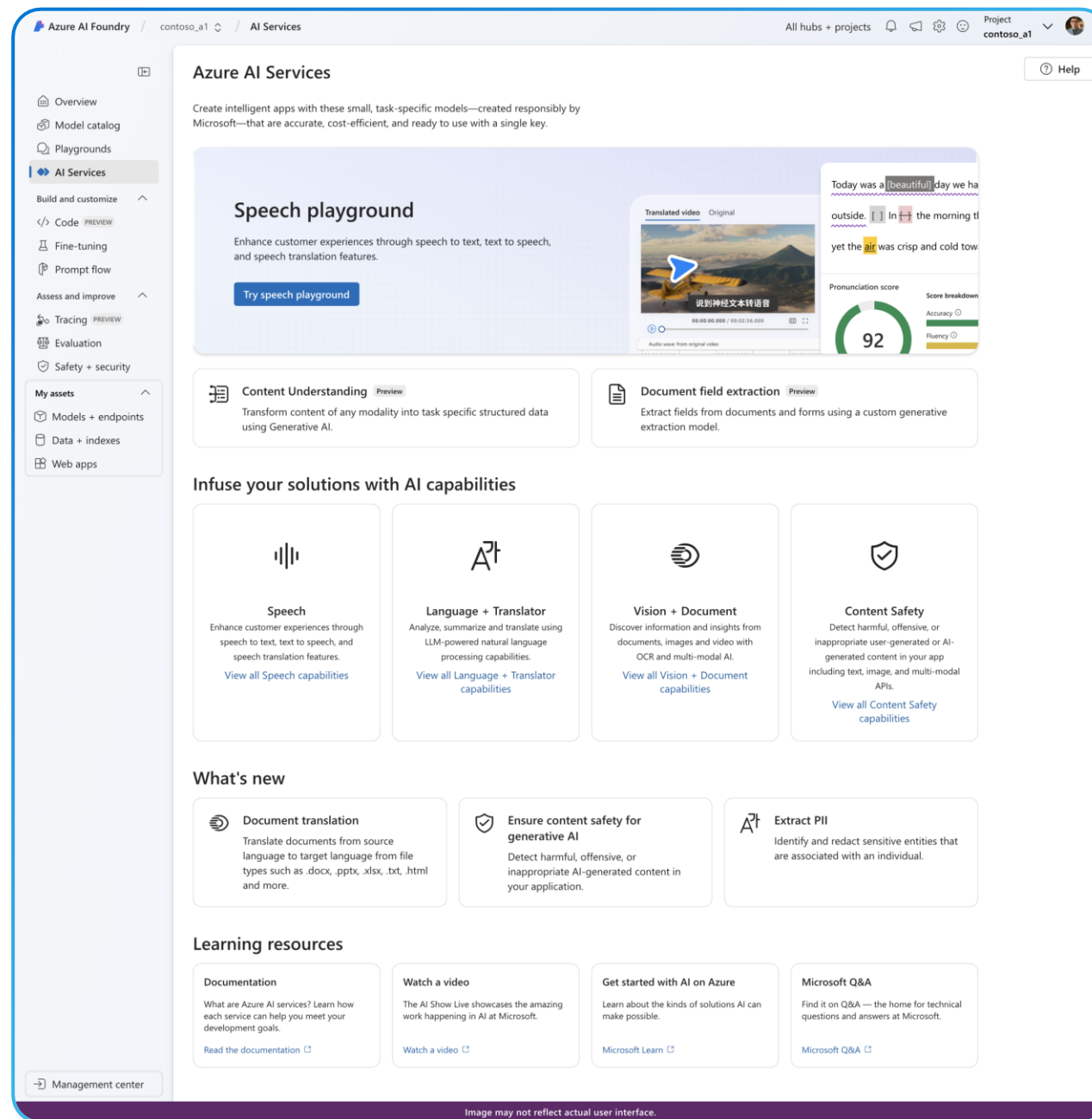
Governance

Monitoring

Observability

Azure AI Services

Build cutting-edge, market-ready AI applications with out-of-the-box and customizable APIs and multi-modal models



Design intelligent apps with Azure AI services

Azure OpenAI Service

- Access to powerful AI models
- Scalable development
- Compliance & security
- Integration with other Azure Services

Azure AI Search

- AI enrichment & semantic ranking
- Generative AI content creation
- Vector search for data organization

Azure AI Speech

- Speech to text (including the Whisper model on Azure OpenAI Service)
- Text to speech
- Speech translation
- Speaker recognition

Azure AI Vision

- Image and face analysis
- Custom model training
- Face detection and recognition
- Document text extraction

Azure AI Content Safety

- AI-driven content moderation for enhanced safety
- Customize safety thresholds for diverse user types
- Detect and prevent Jailbreak Risk from XPIA attacks

Azure AI Document Intelligence

- Automated documentation generation
- Documentation quality analysis
- Interactive documentation experiences
- Natural language understanding for documentation

Azure AI Language

- Task-optimized AI models for text analytics
- Custom industry-specific AI for healthcare
- Custom, industry-specific models

Azure AI Translator

- Multilingual text and speech translation
- Synchronous and asynchronous translation request support
- Native translation of documents and manuals



Azure AI Speech

Azure AI Speech Services

Speech-to-text

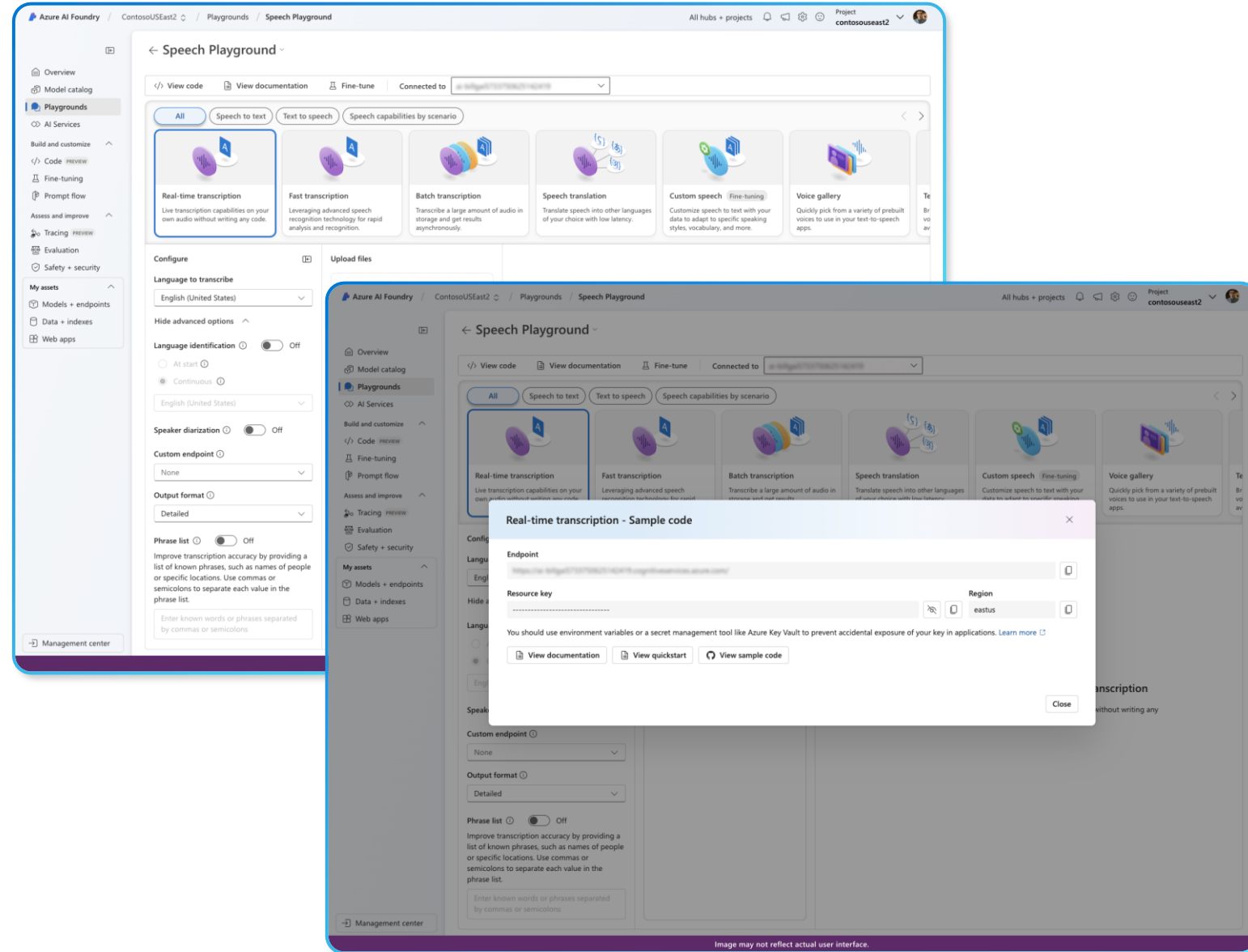
- Contact center – post call analytics and agent assist
- Meeting transcription & insights
- Audio/video transcription & insights
- Dictation
- Caption & subtitles for online and in-person meetings
- Audio and video captions & subtitles

Text-to-Speech

- Context-aware, highly expressive HD voices
- Professional voice/brand voice
- Avatars

Speech Translation

- Interpreter agent in Microsoft Teams
- Video translation API
- Real-time captions and subtitles
- Audio and video caption & subtitle
- Batch video



Available in ai.azure.com

Azure AI Speech-to-text Offerings

Realtime Transcription

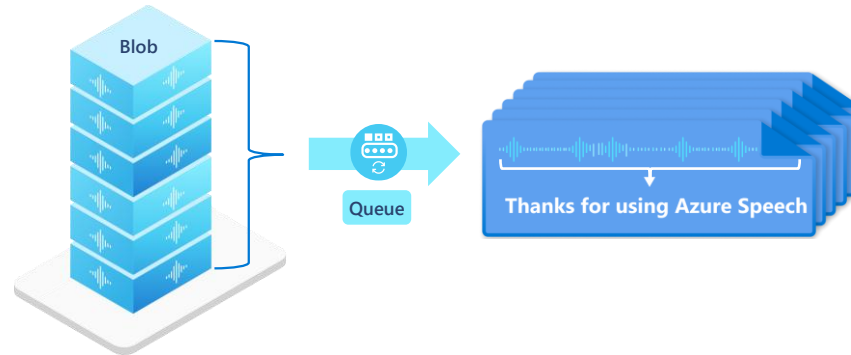
Transcribe from **microphone / streaming audio**, returning transcripts in **real time** with **interim results**.



Realtime Factor ~ 1
First response in a few hundred milliseconds

Batch Transcription

Transcribe **large amount of audio files** in storage, returning transcripts **asynchronously** with **low time urgency**.



Realtime Factor not guaranteed
(Task queueing, low urgency)

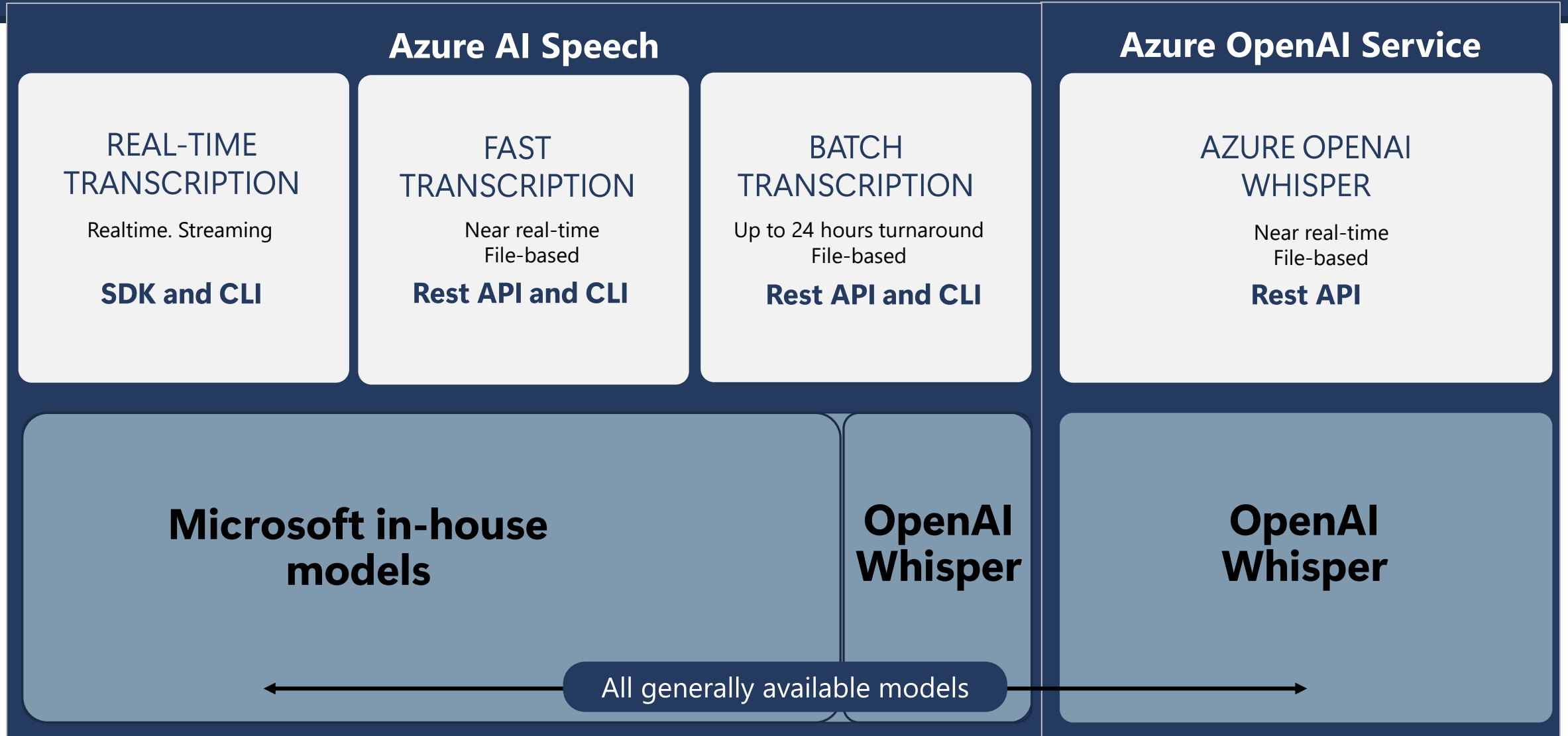
Fast Transcription (new)

Transcribe **one audio file per request**, returning transcripts **synchronously** and **quickly**.



Realtime Factor $\ll 1$
(e.g., 20 mins audio done in 1 min)

Three differentiated speech to text offerings



Azure AI Speech – Fast Transcription

Ultra-fast audio file to text conversion for post call analytics,
voicemail, video editing, and more

Long, context-
aware native display
models

Speaker
diarization

Multi-channel
audio input

Near real-time
response

Azure AI Speech – Fast Transcription

Ultra-fast audio file to text conversion for post call analytics, voicemail, video editing, and more

Long, context-aware native display models

Speaker diarization

Multi-channel audio input

Near real-time response

Azure AI Speech - Custom text-to-speech Avatar

Create a unique and realistic avatar or digital twin using your own video data

Custom avatar model training

Full control of your data,
model, and endpoints

API for video and live chatbot
creation tool

Azure AI Speech – Translation Capabilities

Multi-lingual translation

Automatic language detection

Real-time translated captions

Use text to speech platform voices

Multiple languages in the same session

Video translation

Batch service for e2e video dubbing

Speaker diarization and voice matching

Uses platform voices or personal voice

Human in the loop for content editing

GPT reformulation for translation

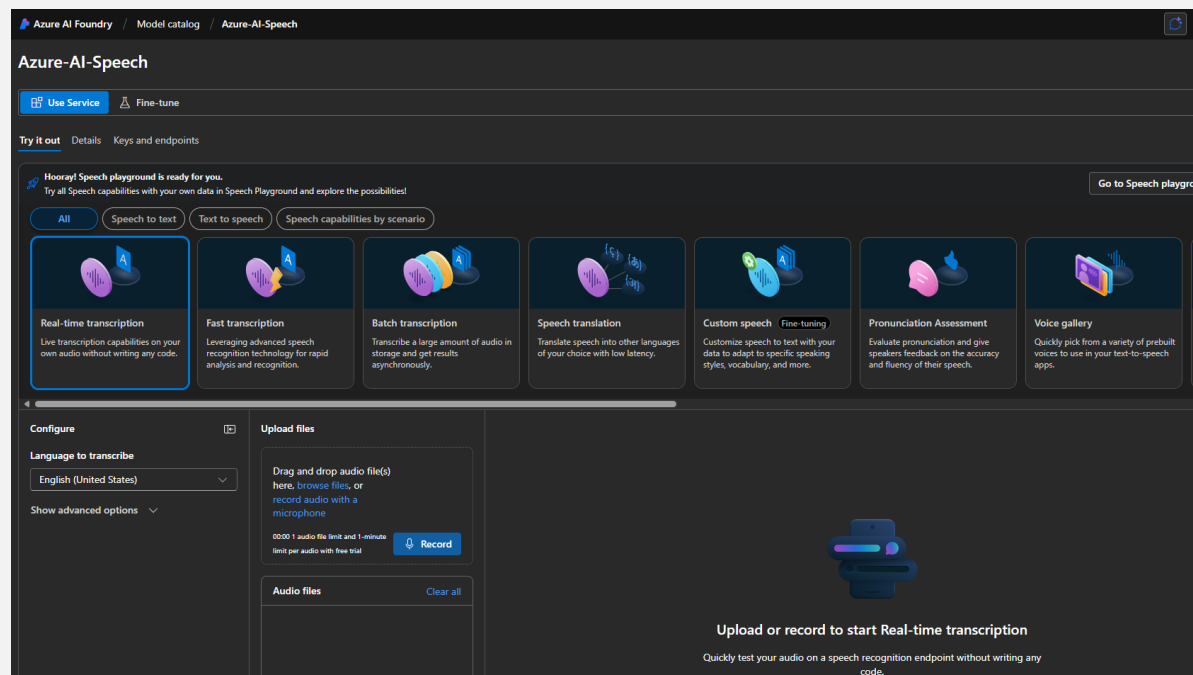
Powering video dubbing in Edge Canary

Azure AI Speech offers transcription technologies for every use-case

	<u>Real-time Transcription</u>	<u>Batch Transcription</u> (include Batch Whisper)	Fast Transcription (new)	<u>Whisper in Azure OpenAI Service</u>
Description	Transcribe live audio from microphone / stream , returning transcripts in real time with interim results .	Transcribe large amount of audio files in storage, returning transcripts asynchronously with low time-urgency .	Transcribe one audio file per request , returning transcripts synchronously and quickly .	A replication of OpenAI Whisper API hosted in Azure.
Input Form	Streaming audio	Audio file (many files via a blob storage URL)	Audio file (one audio file per request)	Audio file (one audio file per request)
Recommended Using Scenarios	- Online Meeting Caption/ Subtitle (translation) - In-person Meeting Caption/ Subtitle (translation) - Contact Center - Voice Agent - Contact Center – Agent Assist (streaming input with partial results) - Voice Assistant/Chatbot - Dictation - Voice Commanding	- Contact Center – Post Call Transcription and Analytics (lower urgency) - Meeting Transcription – Post Meeting Summarization and Analytics (lower urgency) - Recorded Audio/Video transcription (lower urgency)	- Quick Audio / Video edit - Contact Center – Agent Assist (file input) - Contact Center – Post Call Transcription and Analysis (higher urgency) - Meeting Transcription – Post Meeting Summarization and Analytics (higher urgency) - Recorded Audio/Video transcription (higher urgency) - Voice Mail - Video Dubbing	- Offline Audio/Video captioning/subtitling - Audio/Video Transcription - Speech translation to English only
Use if you need	- Streaming input with microphone; - Intermediate recognizing results - Improve quality with Phrase Lists - Improve quality with Custom Speech and deploy endpoints	- Bulk audio files in a storage in a request - Extreme large audio files with >2hrs audio length - Large selection of model versions - Audio formats/codec support in service side - Multiple audio channels - Best quality	- Simple config via one HTTP post - Transcribe much faster than real audio length - One audio file per request - Audio formats/codec support in service side - Good quality.	- Good readability and disfluency - Multilanguage recognition
Don't Use if you need	- Get transcript much faster than real audio time; - Audio formats/codec support in service side	- Streaming input; - Get transcript quickly and synchronously	- Streaming input; - Lexical format result (e.g., "one two three" instead of "123")	- Streaming input; - Lexical format result (e.g., "one two three" instead of "123") - Large audio files - Customization with your data
RTF (Realtime Factor) and Latency	~ 1 (realtime), first-byte latency response in a few hundred milliseconds	Not guaranteed due to task queueing (low time urgency)	<< 1 (e.g., 20 mins audio done in 1 min)	<< 1

Demonstration

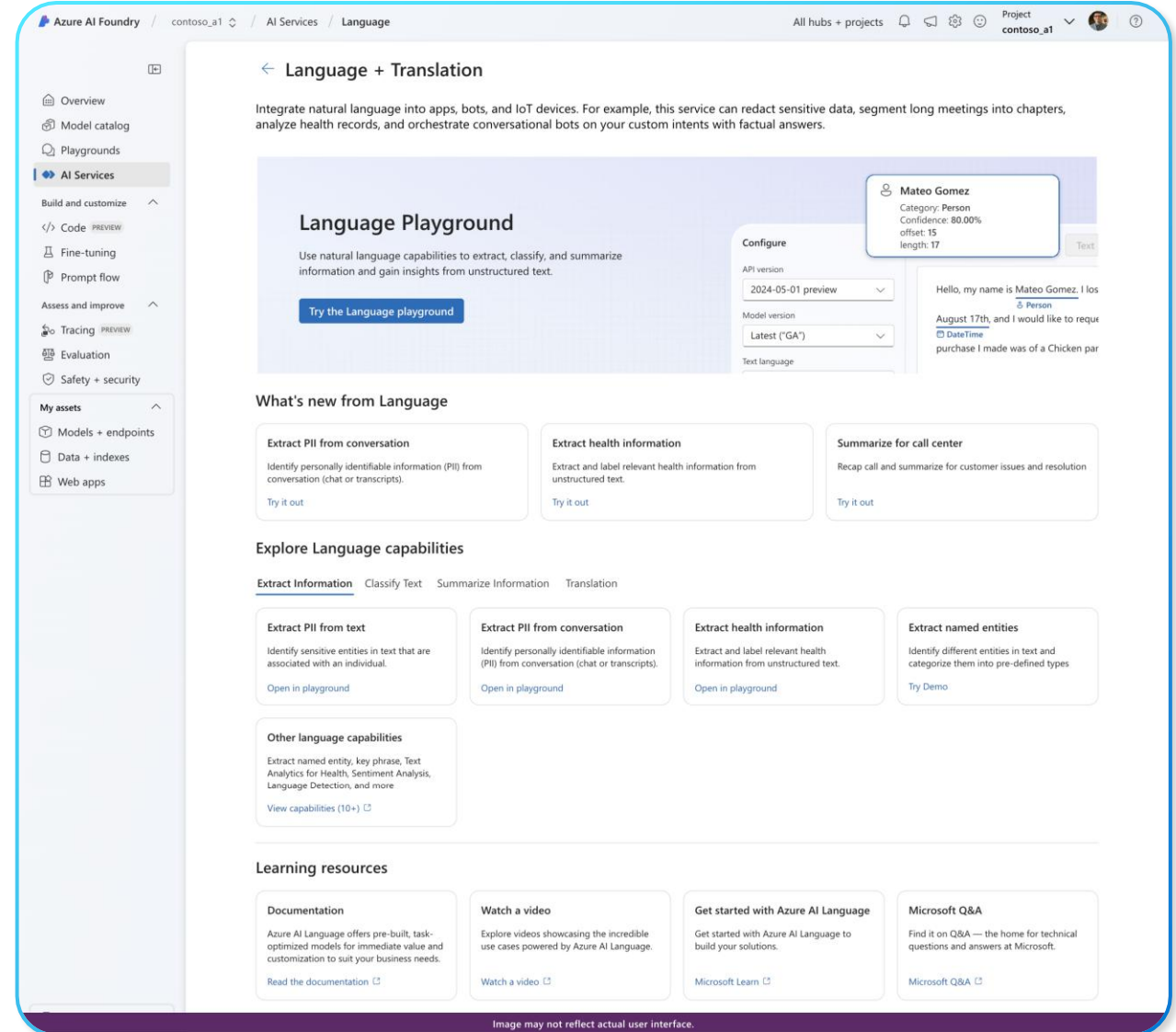
Walk through of the Azure AI Speech Studio in Azure AI Foundry.



Azure AI Language

Azure AI Language and Translation

Build conversational interfaces, summarize documents, and analyze text using prebuilt AI-powered features



Azure AI Language Overview

- A **unified service** with one Azure resource, one user experience, one set of APIs
- **Multilingual** for +100 languages via Z-Code and Z-Code++.
- Powered by large language models
- **Pre-built**, task-optimized models for immediate value
- **Customization** to adapt models to customer specific needs
- **Flexible deployment** through Azure hosted service or as customer-hosted containers

Summarization



Document summarization



Conversation summarization



Custom Summarization

Conversation AI



Question answering



Custom Question Answering



Conversational language understanding

Text Analytics

Extract information



Extract NER and PII



Extract key phrases



Find linked entities



Extract health information



Custom Health



Custom NER

Classify text



Analyze sentiment and mine opinions



Detect language



Custom Classification



Custom Sentiment

Regulatory Compliance



CSA



FedRAMP



GDPR



HIPAA



HITRUST



ISO



PCI



SOC



US DoD

Personally Identifiable Information (PII) Detection

- Identifies, categorizes, and redacts sensitive info in unstructured text
- Supports a wide range of PII info, including region-specific PII, PHI and PCI
- Supports conversational scenario with a specialized model (aka. Conversation PII)
- 79 languages supported

Typical use cases:

- Mask sensitive info to avoid the leak, inappropriate handling or data bias
- Auto label and classify docs based on the sensitivity

Sample text: The food is great, and the place was impeccably clean. You can order from their online menu at www.contosohouse.com, call 312-555-0176 or send email to order@contosohouse.com!	Recognized entity categories: URL PhoneNumber Email
Redacted text: <i>The food is great, and the place was impeccably clean. You can order from their online menu at www.contosohouse.com, call ***** or send email to *****!",</i>	

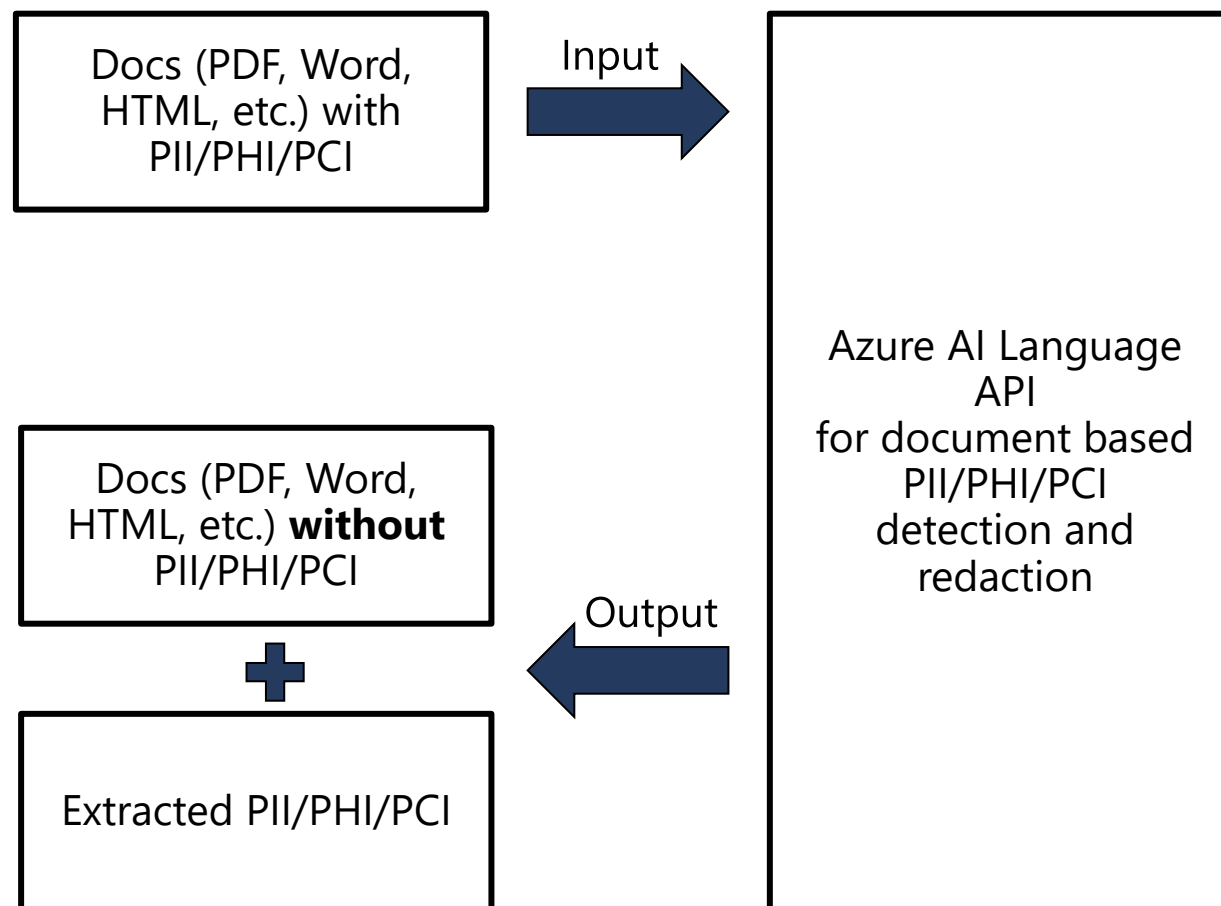
PII Detection – Supported Categories

Basic PII Info	Government & Country/Region-Specific	Azure Information
Person		Azure DocumentDB Auth Key
Person Type	U.S. Social Security Number (SSN)	Azure IAAS DB Connection String
Phone Number	U.S. Driver's License Number	Azure SQL Connection String
Organization – <i>incl. types:</i>	U.S. Passport Number	Azure IoT Connection String
• Medical	U.S. Individual Taxpayer Identification Number (ITIN)	Azure Publish Setting Password
• Stock Exchange	U.S. Drug Enforcement Agency (DEA) Number	Azure Redis Cache Connection String
• Sports	U.S. Bank Account Number	Azure SAS
Address	+ <i>more than 90 PII entities across 31 other countries/regions</i>	Azure Service Bus Connection String
Email		Azure Storage Account Key
URL		SQL Server Connection String
IP Address		
Date Time – <i>incl. type:</i>		
• Date		
Quantity – <i>incl. type:</i>		
• Age		

PII Redaction on Native Document Formats (preview)

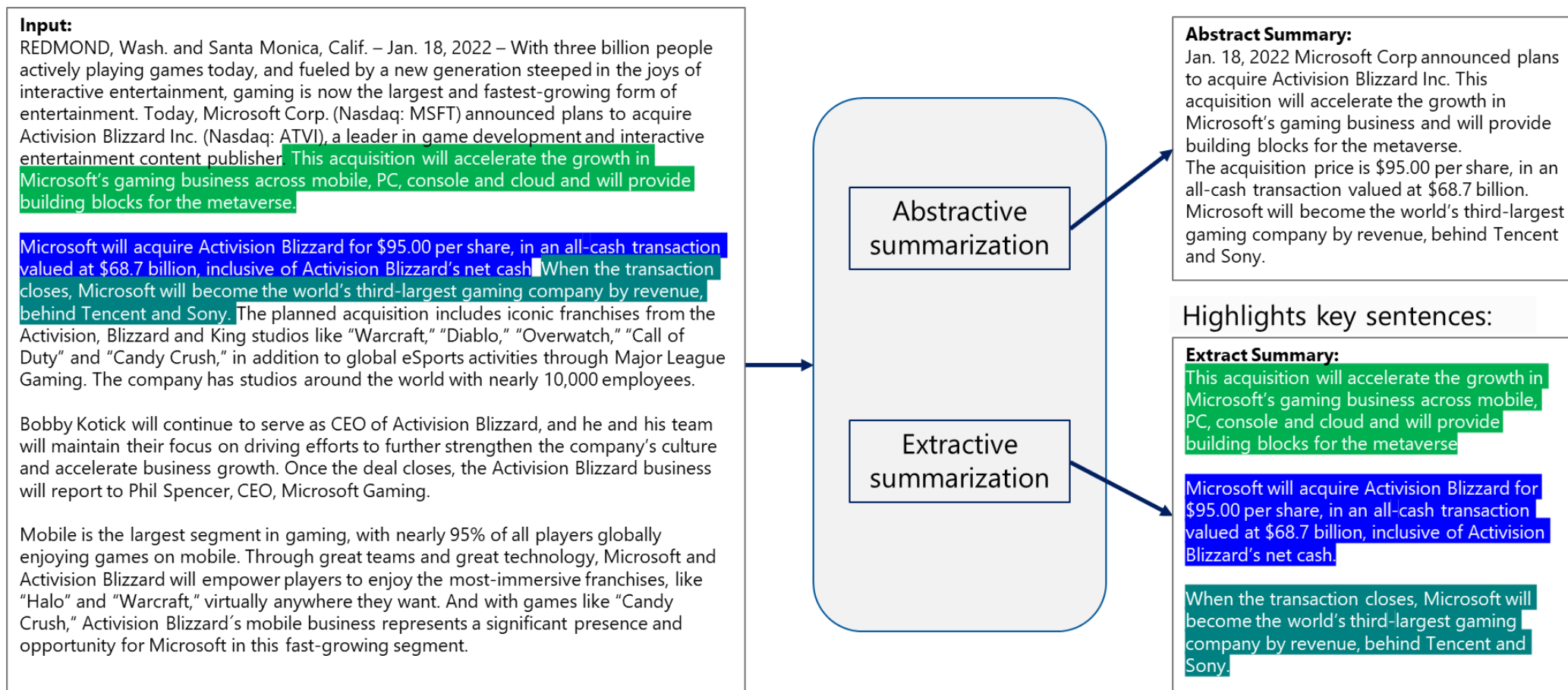
Identifies and redacts sensitive PII info directly in native documents –
No more doc pre-processing and post-processing needed!

PII entities and values extracted from documents are included in the API response payload, which can be stored separated and used for later



Document Summarization

Auto-generate a concise summary from both abstractive and extractive approaches to improve the productivity of information analytics and processing.



Named Entity Recognition (NER)

Identifies and categorizes entities in unstructured text
30+ pre-defined entity categories

(includes subcategories)
79 languages supported

Typical use cases:

Search indexing optimization

Business process automation
based on entity categories and values

Customer analysis to highlight relevant topics

Sample text:

As dozens of **heads of state** convene for the annual **U.N. General Assembly** in **New York** **this week**, the lingering conflict in **Syria** is taking a back seat while tensions in the **Persian Gulf** and global trade **wars** take center stage. **Now** in its **ninth** year, many **Syrians** fear the unresolved **war** has become a footnote in a long list of world crises, with weary **leaders** resigned to live with President **Bashar Assad** ruling over a wrecked and divided country for the foreseeable future. On the eve of the global gathering in **New York**, **U.N. Secretary-General Antonio Guterres** announced that a long-awaited committee that would draft a new Syrian constitution has been finalized — a step the **U.N.** hopes will put the war-ravaged country on track for a political solution.

Recognized entity categories:

PersonName

PersonType

Location

Organization

DateTime

Quantity

Event

Available in containers

Mines text for clues about positive or negative sentiment
Associate sentiment with specific aspects of the text (i.e., targets & their assessment)
90+ languages supported

Mines text for clues about positive or negative sentiment
Associate sentiment with specific aspects of the text (i.e., targets & their assessment)
90+ languages supported

What people think about Barcelona?



Polarity



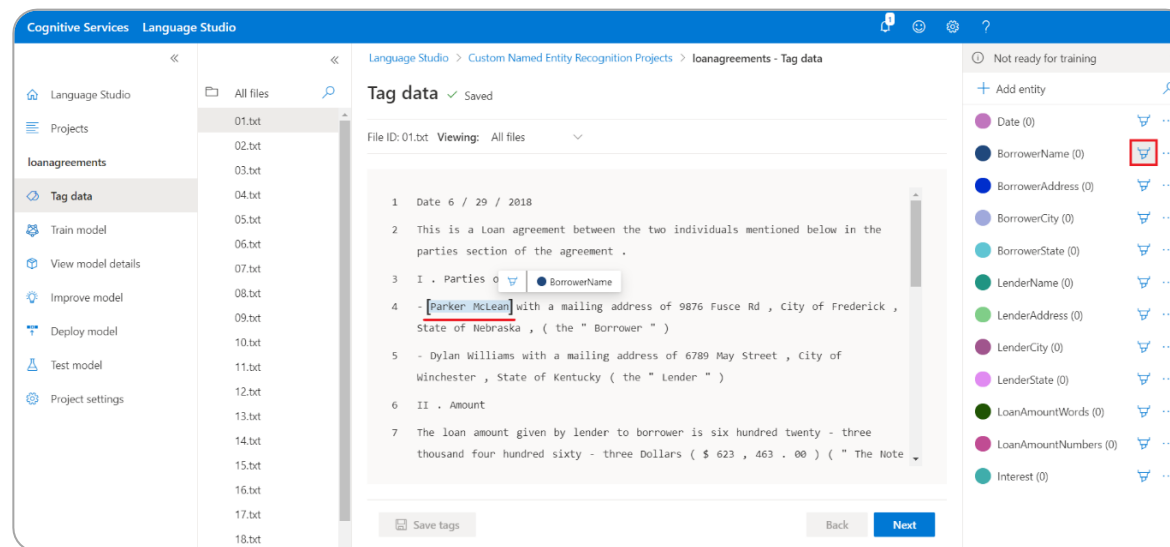
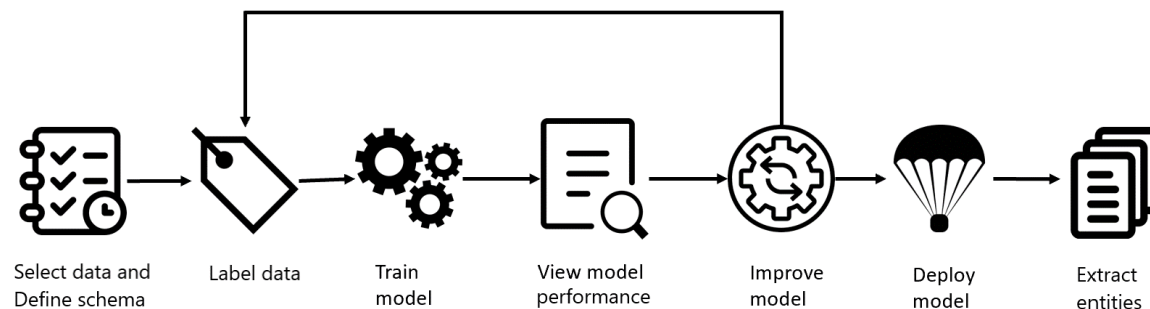
Count of Id by NegatedValue



Custom Named Entity Recognition

Your own custom models to extract domain-specific entities from unstructured text
95+ languages supported

Iteratively label data, train, evaluate, and improve model performance in Language Studio before making it available for consumption



Key Phrase Extraction

Extracts key words and phrases
Powered by transformer-based model

Tailored for both short texts
(without much topic information) and long texts
(where it's easier to track topic)
Supports 90+ languages

Typical use cases:
Search indexing optimization
Article tags for topic filtering
World cloud

Sample text:

Around a third of the global population in the world's poorest countries have been pinning their hopes on India to deliver their Covid-19 vaccines. The Covid-19 outbreak is sad. Then the virus overwhelmed India itself. The outbreak has caused widespread misery for the country's population of 1.3 billion. Hospitals are overwhelmed, oxygen is scarce and the death toll of 200,000 is thought to be a huge undercount. Covid-19 is a dangerous disease.

Extracted key phrases:

The Covid-19 outbreak, Covid-19 vaccines, poorest countries, widespread misery, death toll, huge undercount, dangerous disease, global population, third, world, hopes, India, virus, country, Hospitals, oxygen

Conversational Language Understanding (CLU)

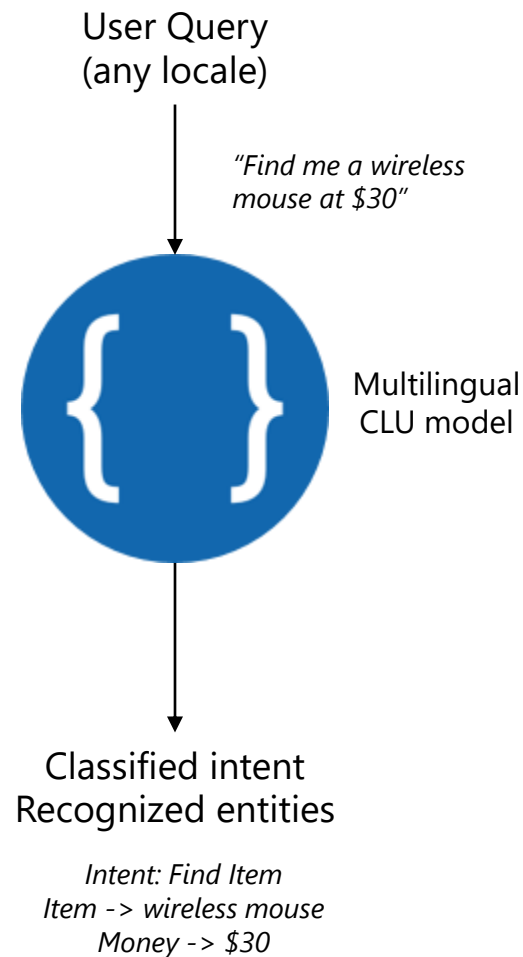
Train and deploy your custom CLU model to predict intents and extract entities using Language Studio

Understands the semantics meaning, not just text patterns

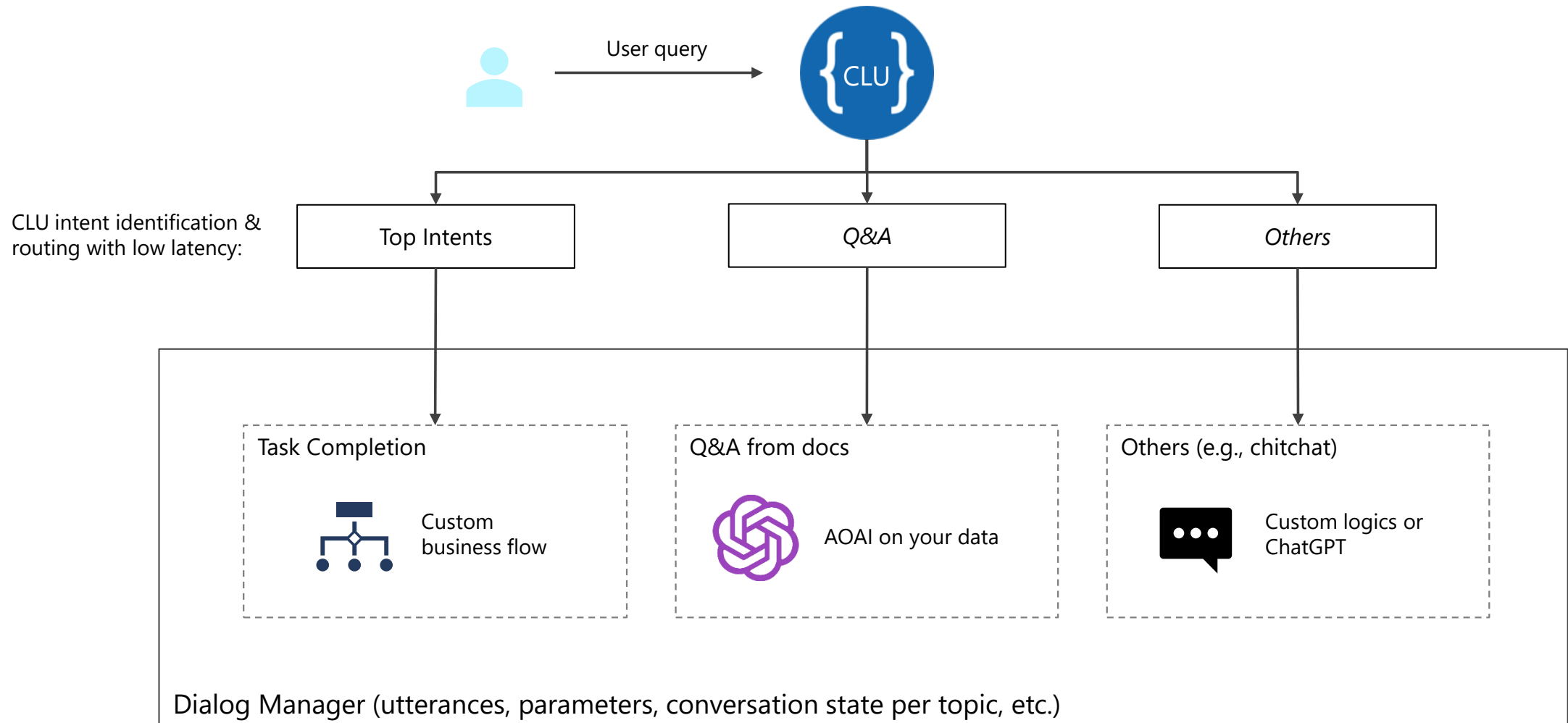
Native multilingual support: Train in one language, predict in all other 96+ languages

Azure OpenAI-powered data labeling to accelerate the custom model building

Orchestrate with Azure OpenAI to enable your conversational AI experience with higher quality, lower cost and latency



Orchestrate CLU with Azure OpenAI Service



Translator API overview

Translate Text

Into multiple languages
Transliterate from one script to another

Translate Documents

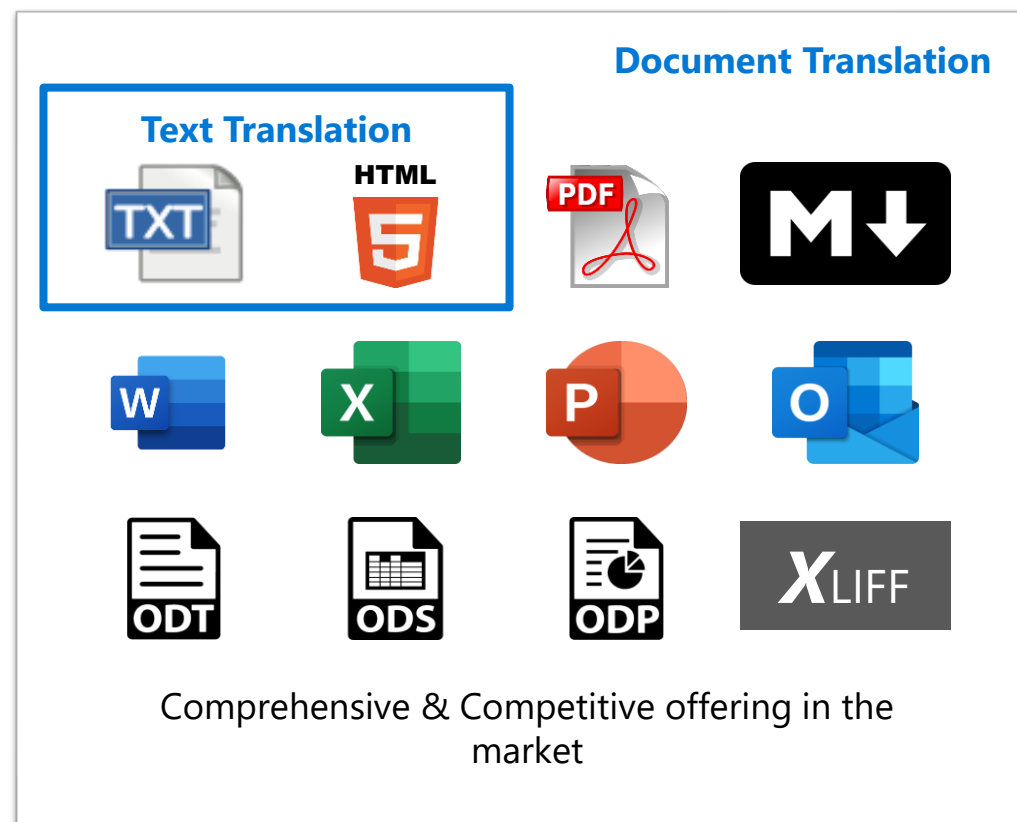
Retaining structure and layout
Into multiple languages
Large volumes of text data
Real time or batch processing for large documents

Custom Models

Build custom models with easy-to-use
Custom Translator
Uses general & custom models to
translate Text and Documents

Security and Compliance

Enterprise ready with Azure AD
authentication & Managed Identity



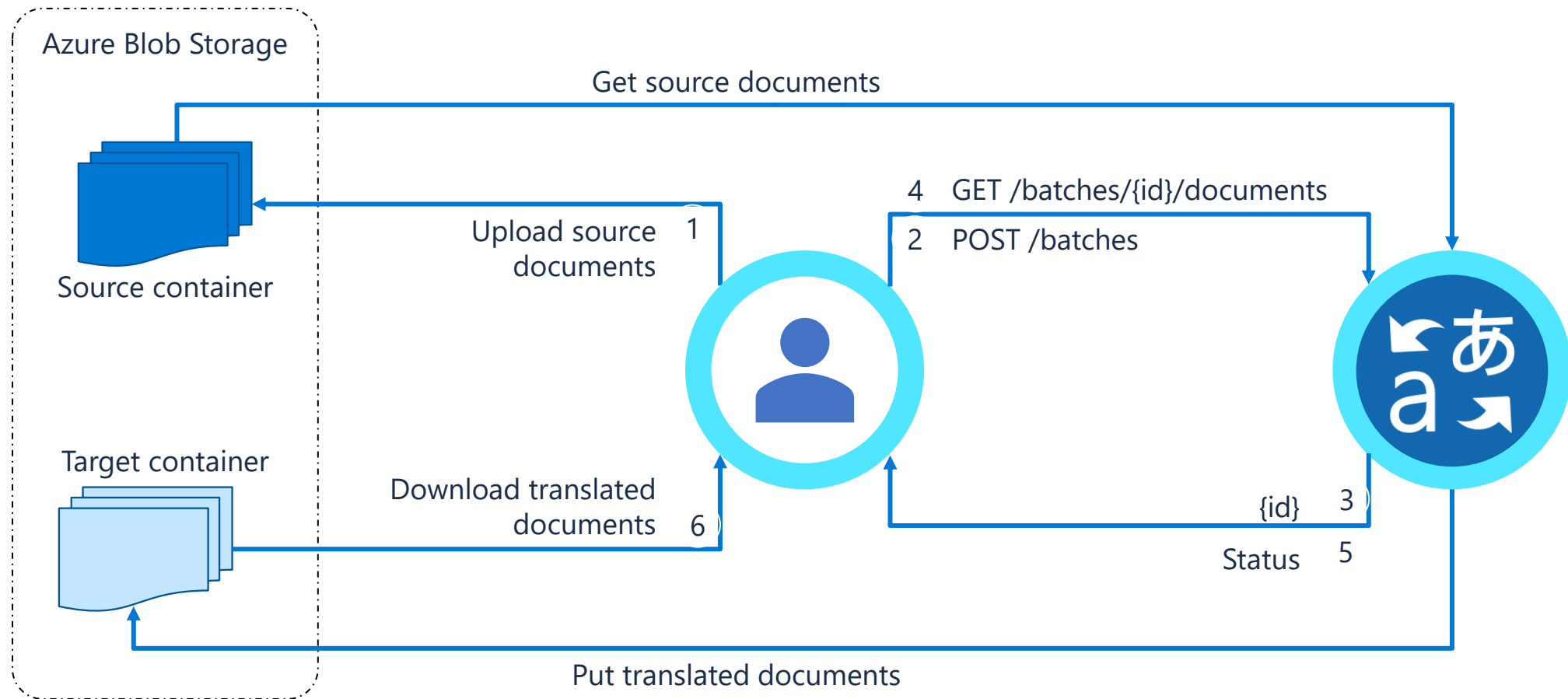
Document Translation overview

- Translates rich documents retaining structure and layout
- Translates documents into multiple languages
- Translates batch of large documents
- Translates multi-lingual documents
- Uses general & custom models
- Applies custom glossaries
- Azure AD authentication
- Secured access to storage using managed identity

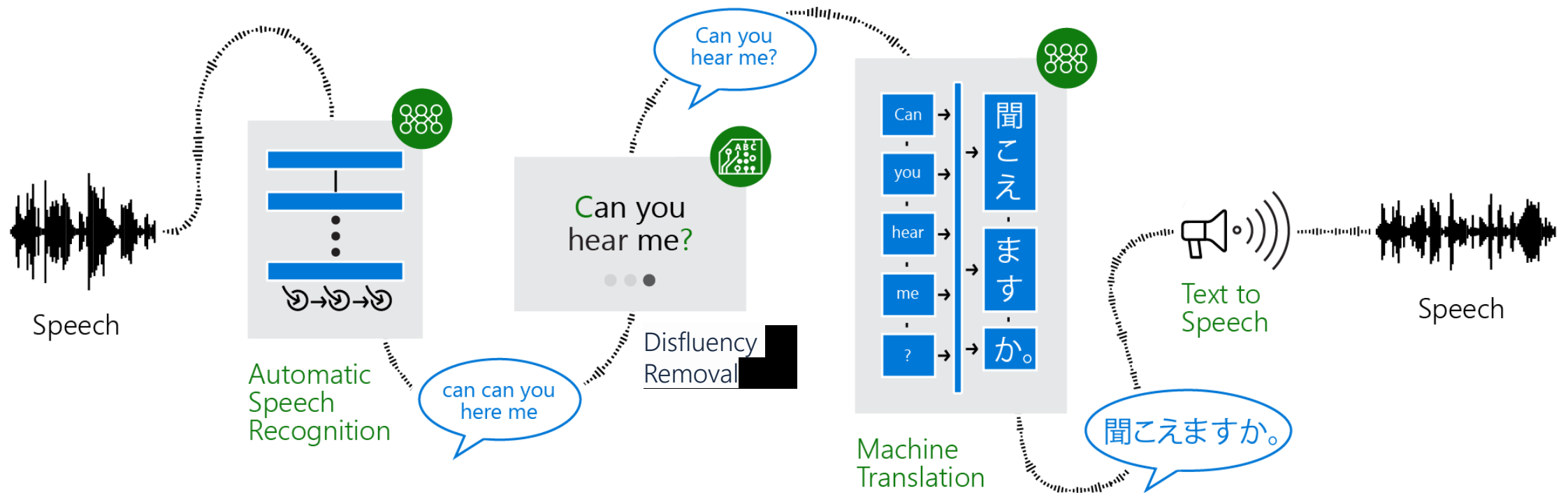


Comprehensive & Competitive offering in the market covering all popular document formats

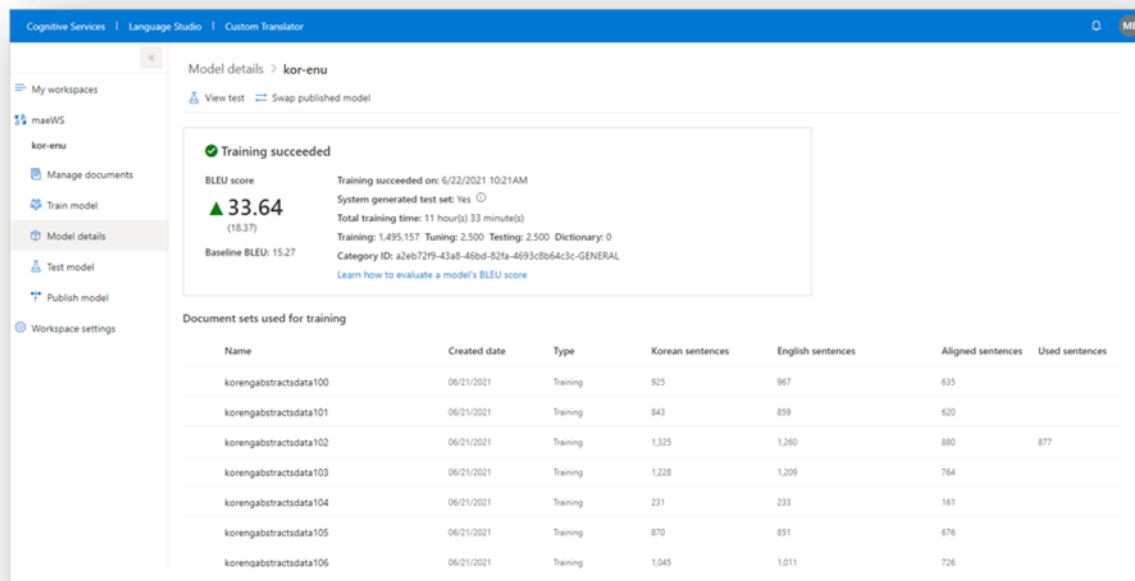
Document Translation simple flow



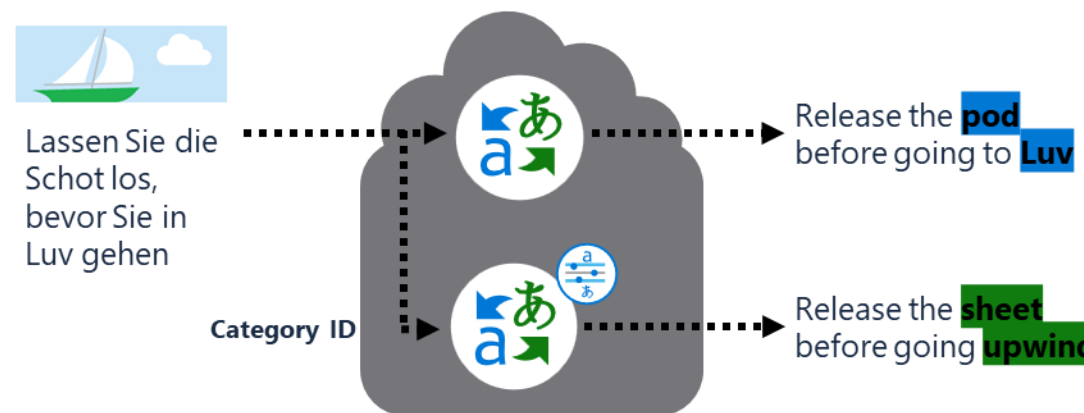
Speech Translation SDK



Using Custom Translator



All popular document formats



1

Upload
your language training
data

2

Train
your new custom
model

3

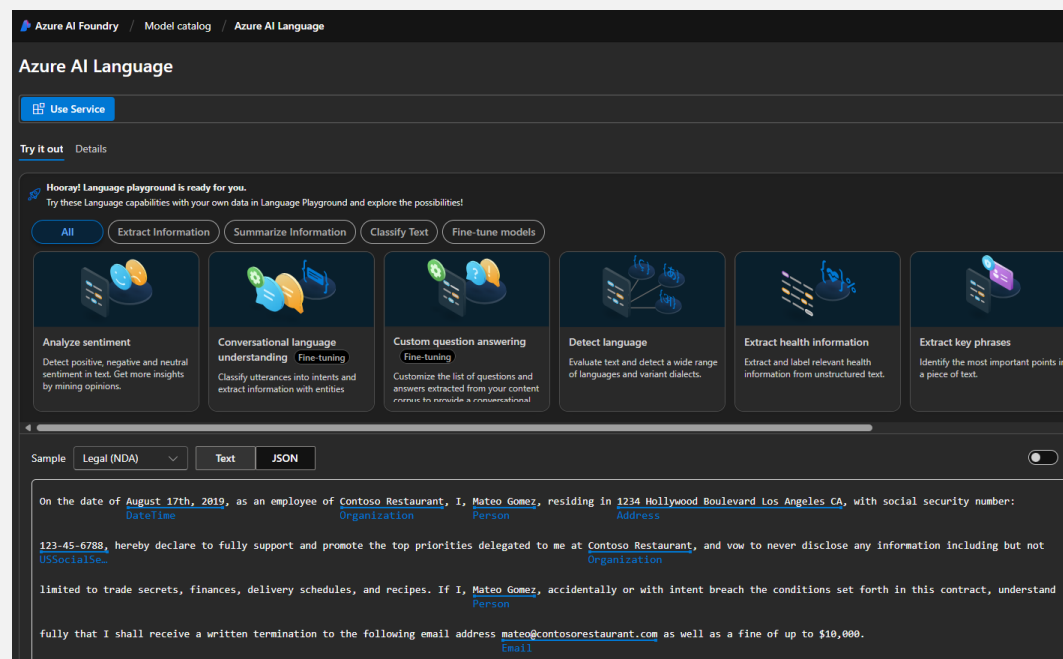
Test
to compare your new
custom translations to
standard translations

4

Deploy
to translate live via
Translator REST API

Demonstration

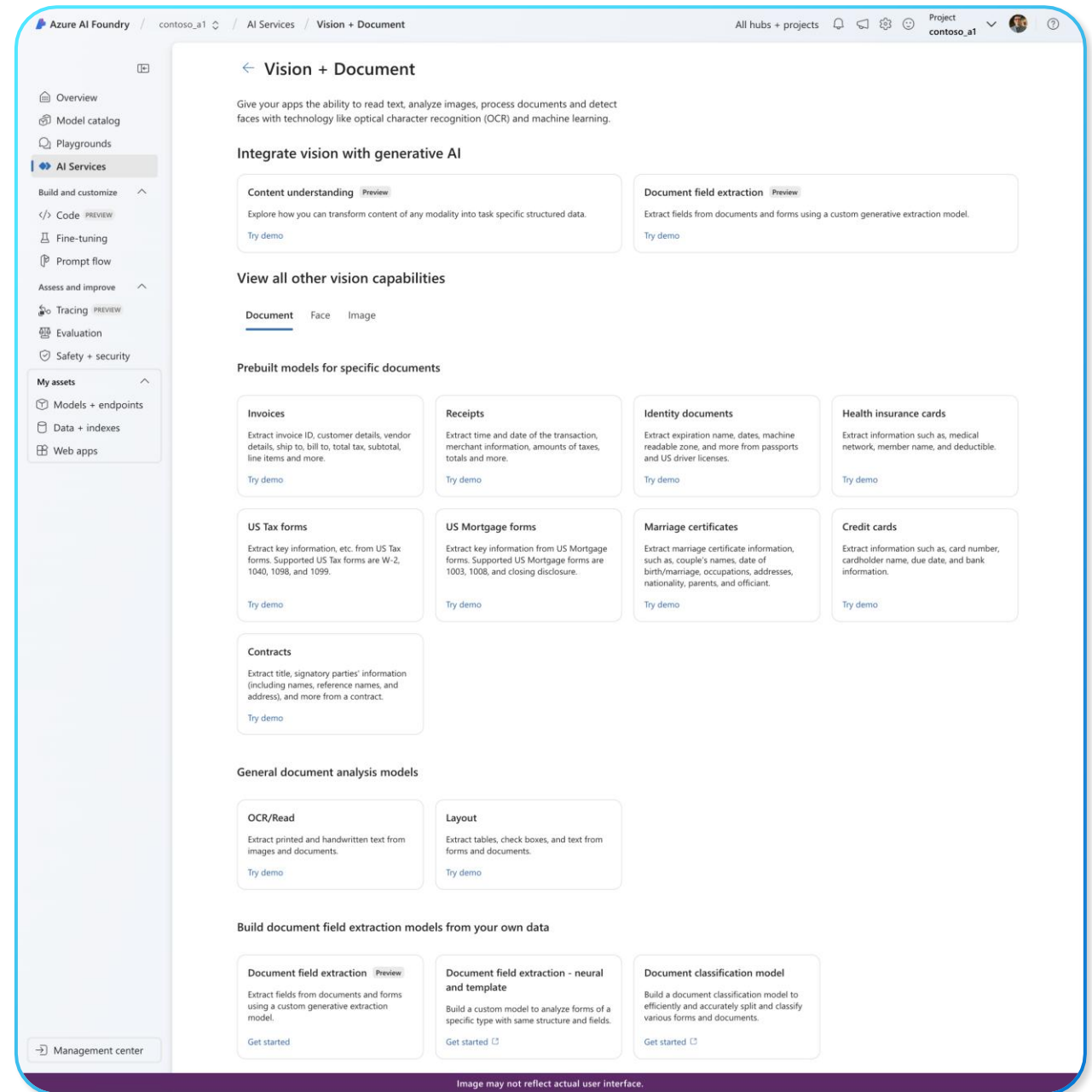
Walk through of the Azure AI Language Studio in Azure AI Foundry.



Azure AI Vision / Content Understanding

Azure AI Vision and Content Understanding

Read text, analyze images, and detect faces with OCR and machine learning



Azure AI Vision

GPT-4 TURBO ENHANCEMENTS

Video prompting

Object grounding

Dense text accuracy

OCR

FACE

IMAGE ANALYSIS

SPATIAL ANALYSIS

Printed Text Extraction

Detection

Tagging & Captioning

Person Tracking

HIGH QUALITY PRE-TRAINED MODELS

Handwritten Extraction

Recognition & Liveness

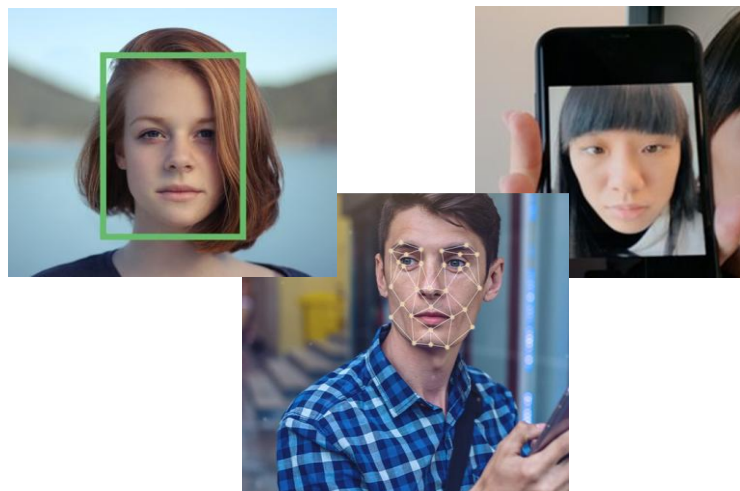
Content Moderation

Object Tracking

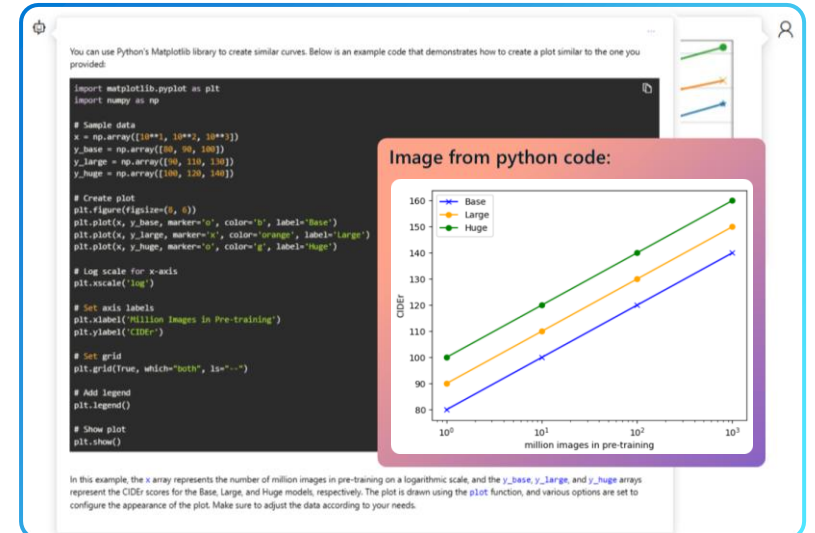


You are an AI assistant which generates listings for vacation rentals. Generate exciting content for this image but don't talk about content that you cannot see. Follow the format of an attention grabbing title and provide a description that is only 6 sentences long.

Analyze objects in images and videos and provide detailed descriptions



Detect and recognize facial liveness



Convert images into code

Azure AI Content Understanding

Build with documents, audio, images, video and text

Streamline multi-modal
app development

Task specific structured
output for agentic workflows

Pre-built templates
to speed development

Azure AI Content Understanding

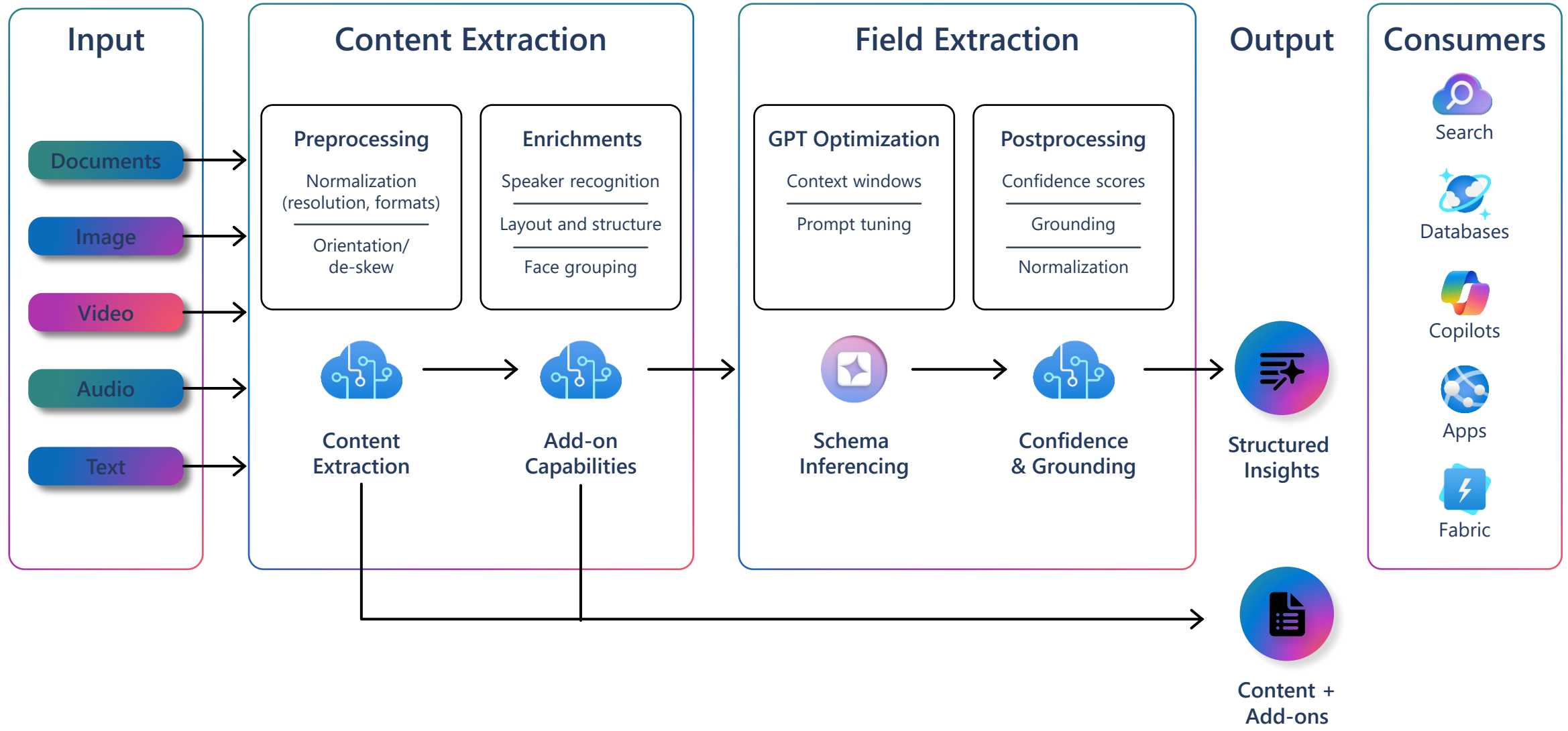
Generate insights from documents, text, audio, images, and video

Differentiators:

- Streamline multi-modal app development
- Task specific structured output for agentic workflows
- Pre-built templates to speed development



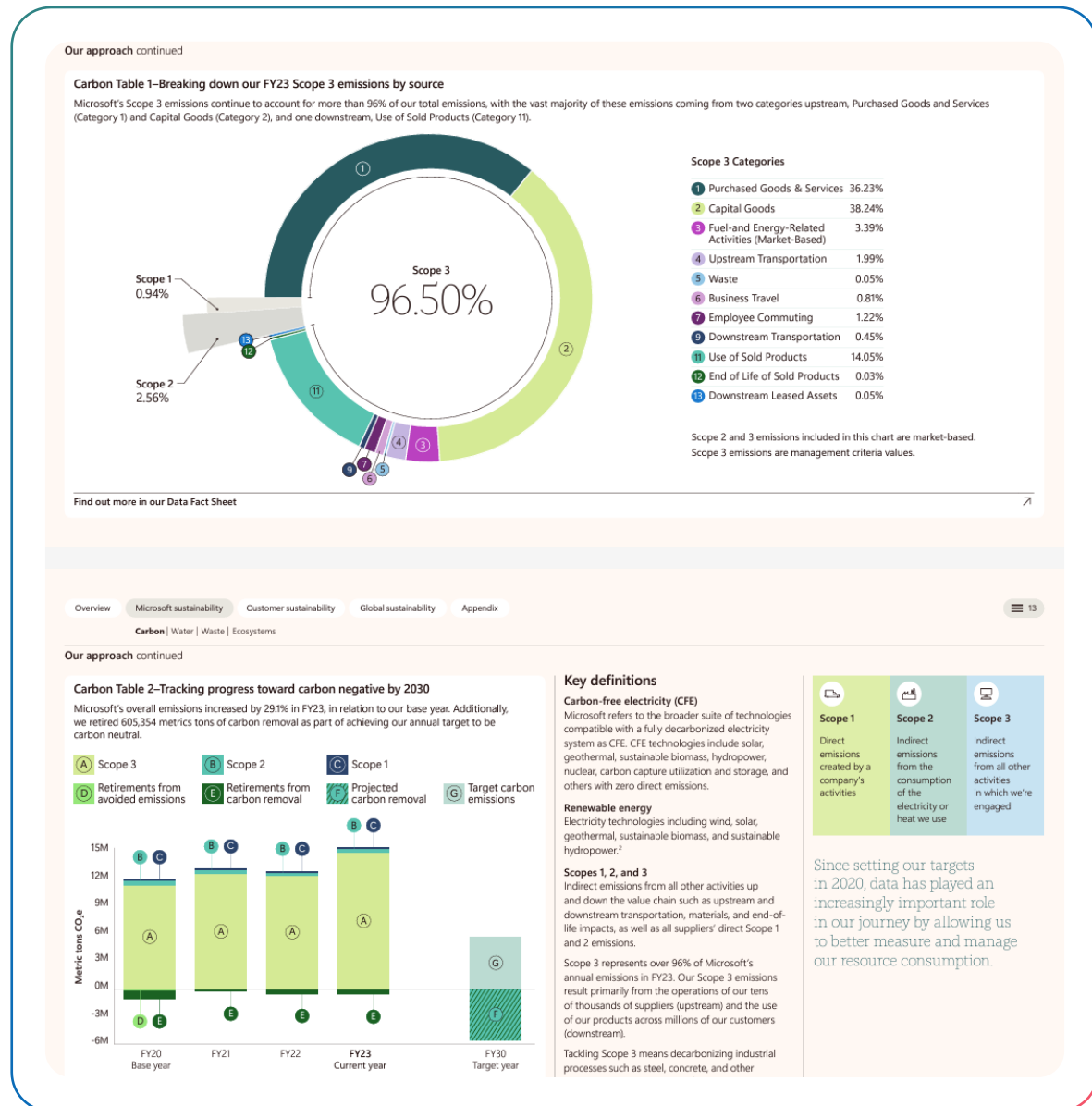
Components of Content Understanding



Content Understanding

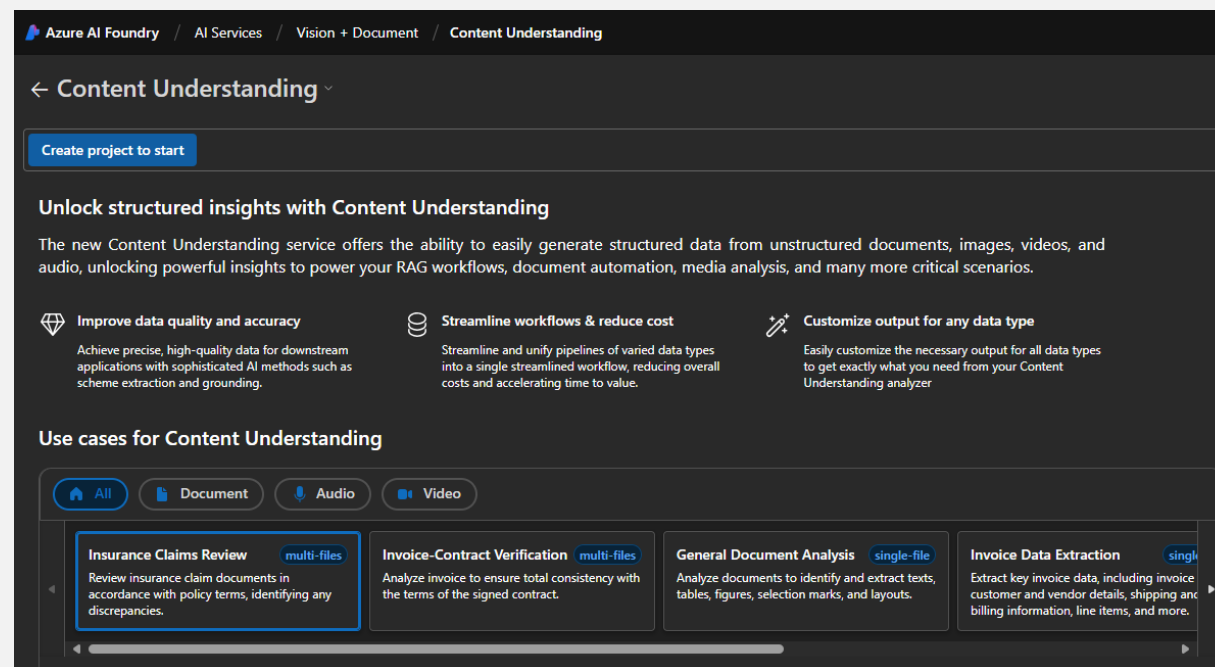
Understanding content from tables, charts and graphs

Content extraction for Document chat with Azure AI Search



Demonstration

Walk through of the Azure AI Vision /Content Understanding in Azure AI Foundry.



<https://ai.azure.com/explore/aishervices/vision>

Azure AI Search

Azure AI Search



Azure AI Search

**Battle-tested
retrieval**

**Streamlined data
pipeline**

**Seamless dev
experiences**

**Enterprise-ready
foundation**

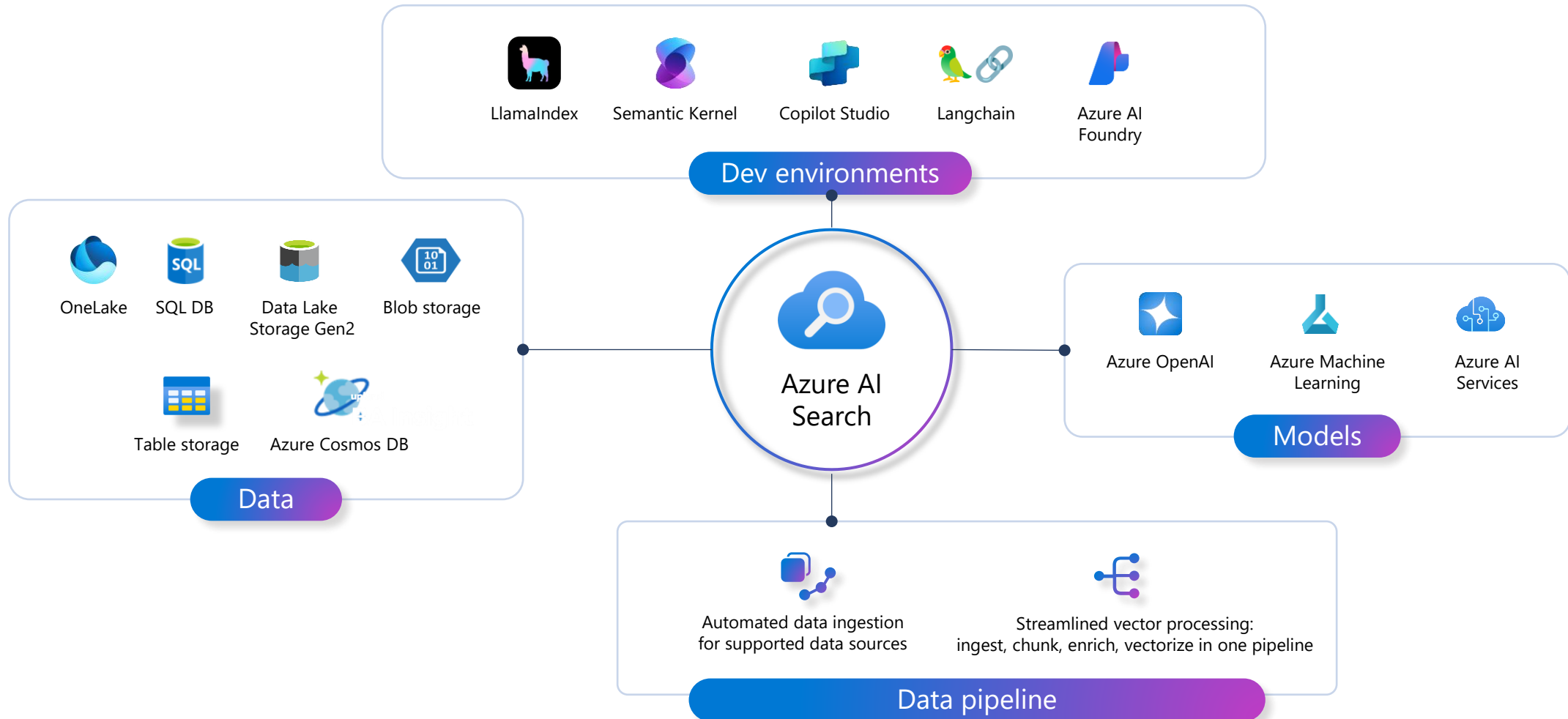
Azure AI Search

Retrieval system
built for scale

- Full-featured
- Create embeddings using any model
- Explicit and transparent vector data processing
- Exhaustive KNN search & ANN search
- Multi-vector



Seamless integrations for your GenAI deployments



Battle-tested retrieval

Vector Search (ANN, KNN)	✓	Query rewriting	✓
Hybrid Search (RRF)	✓	Multi-vector search	✓
Search reranking (BM25)	✓	Multi-lingual search	✓
Exact keyword match	✓	Geospatial search	✓
Facets	✓	Auto-complete	✓

Enterprise-ready Platform



**Data
encryption**



Authentication



**Network
security**

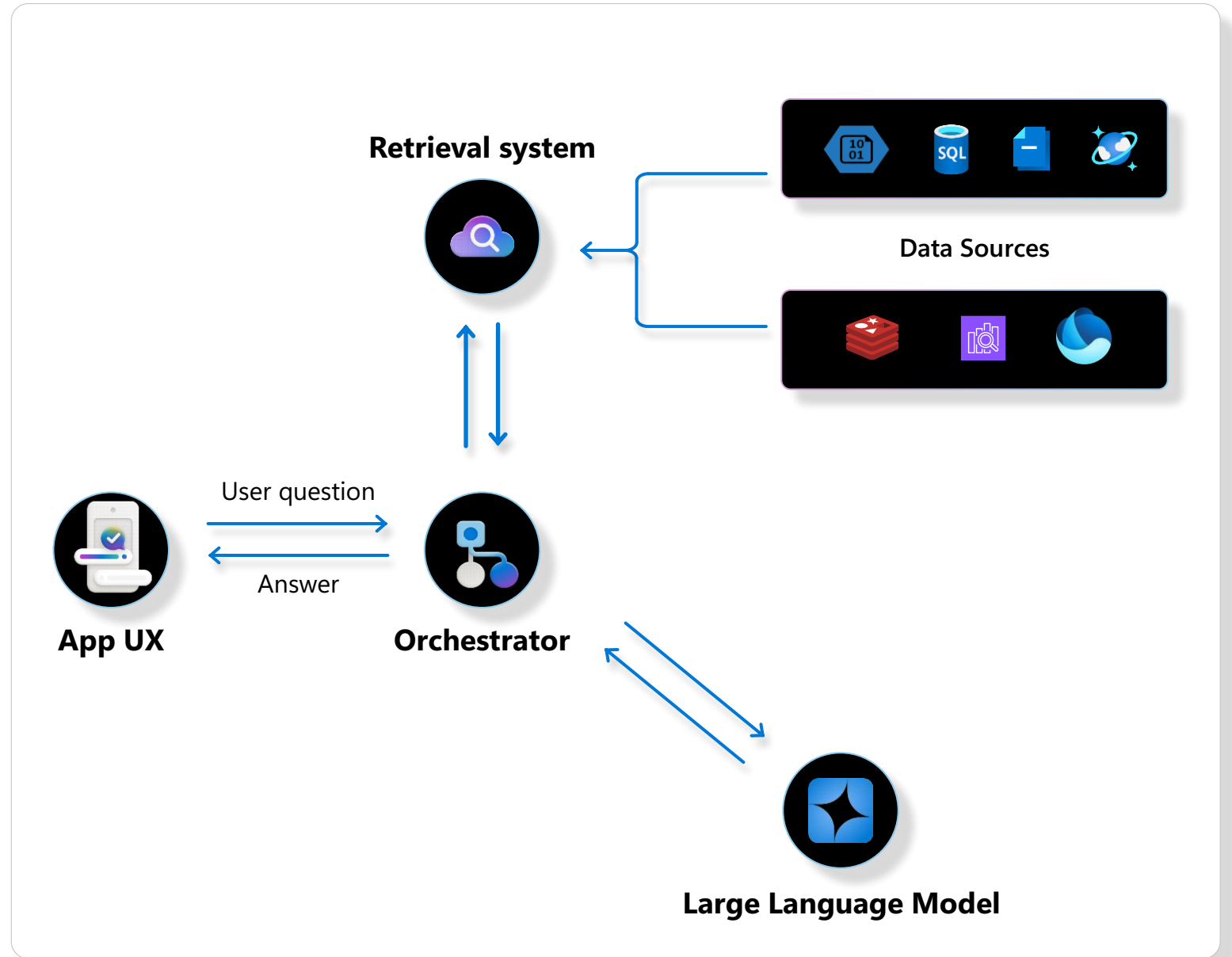


**Data
privacy**

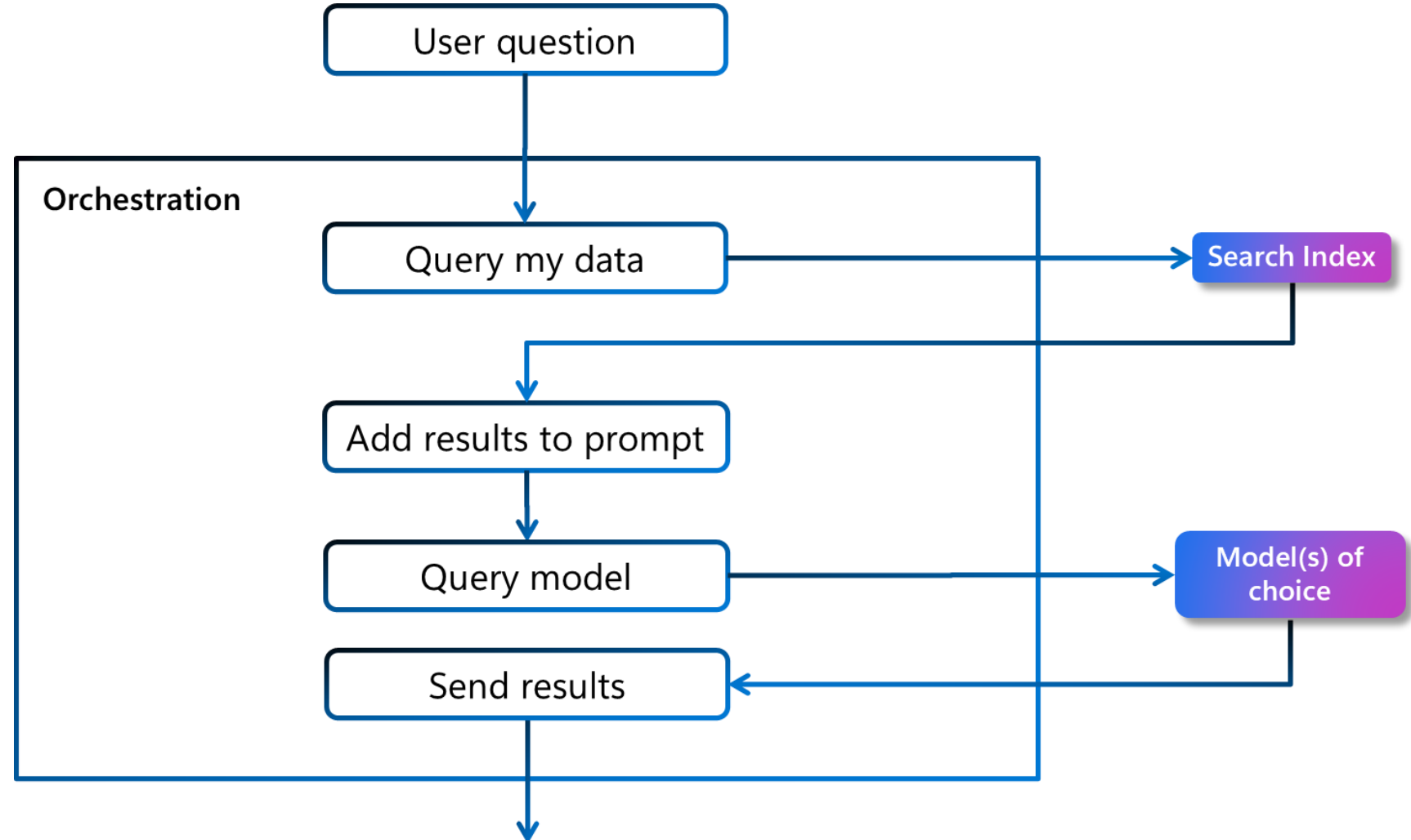


Compliance

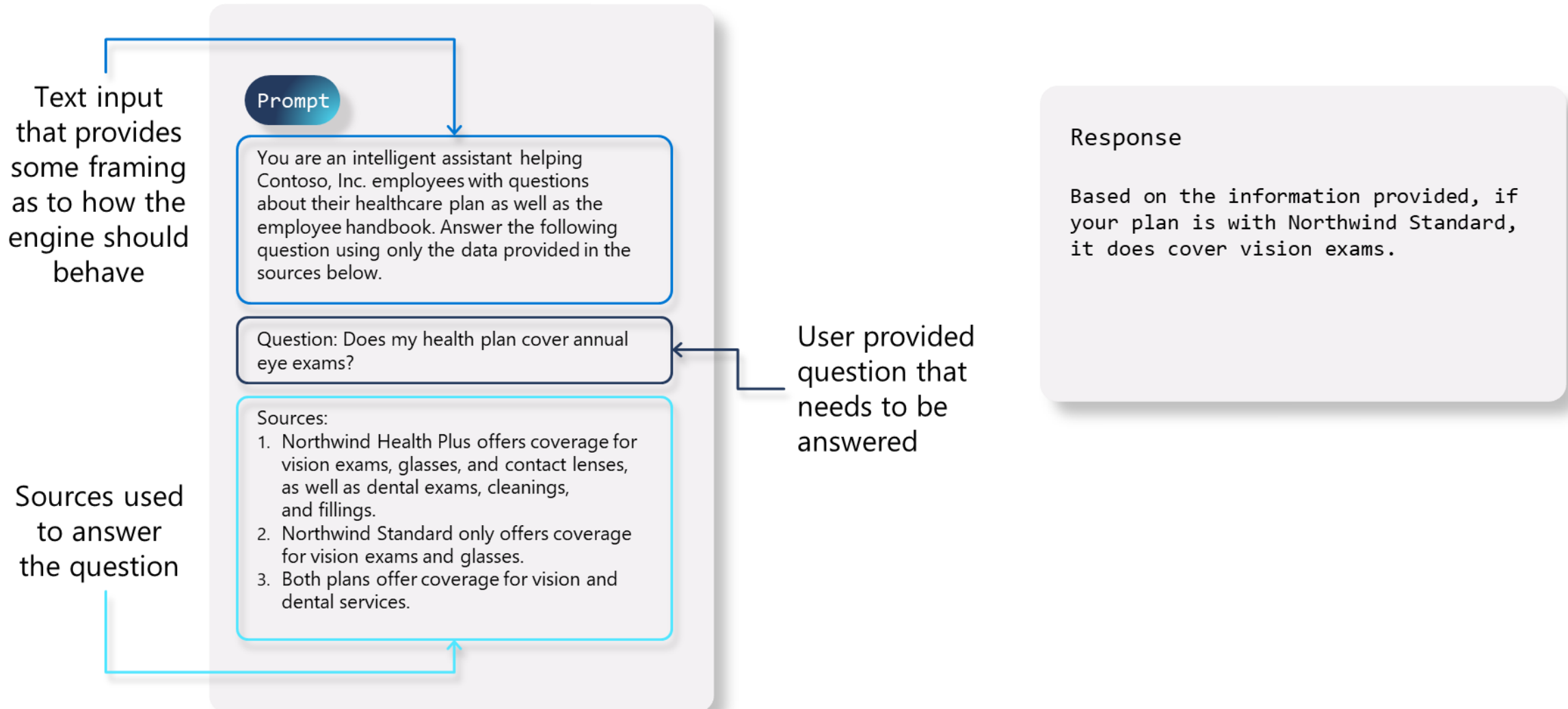
Retrieval-augmented generation



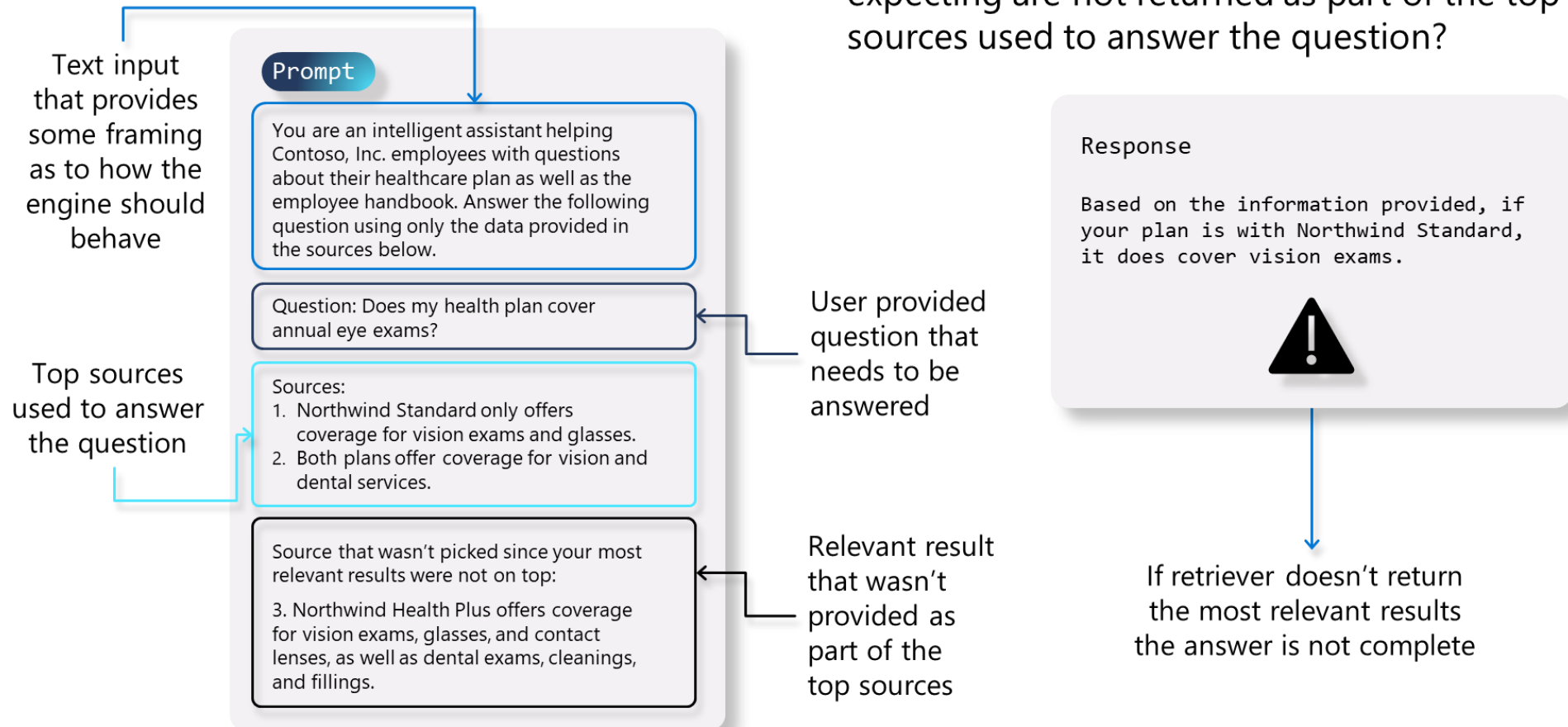
RAG query pipeline



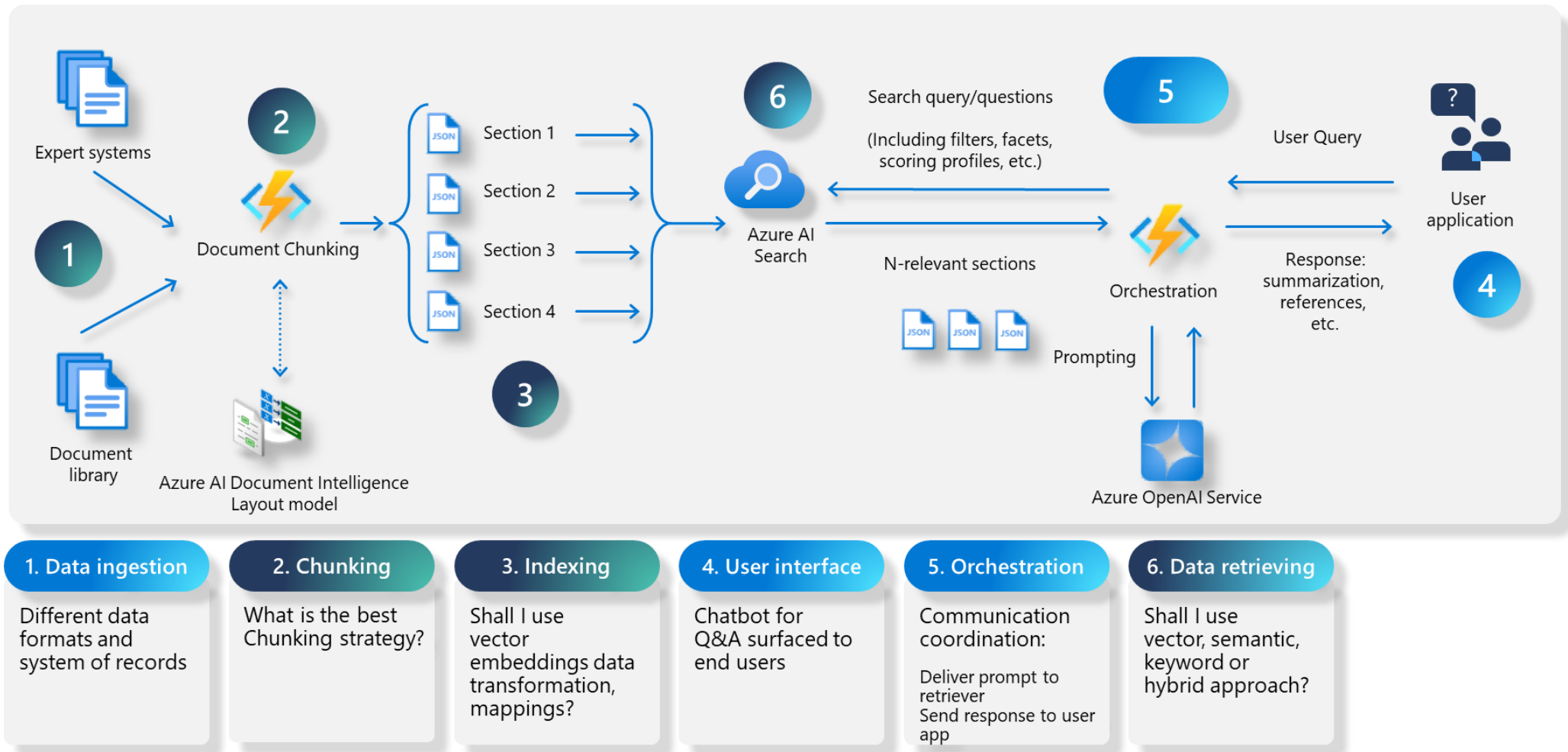
RAG example: bring your data to the prompt



Why is retrieval important?



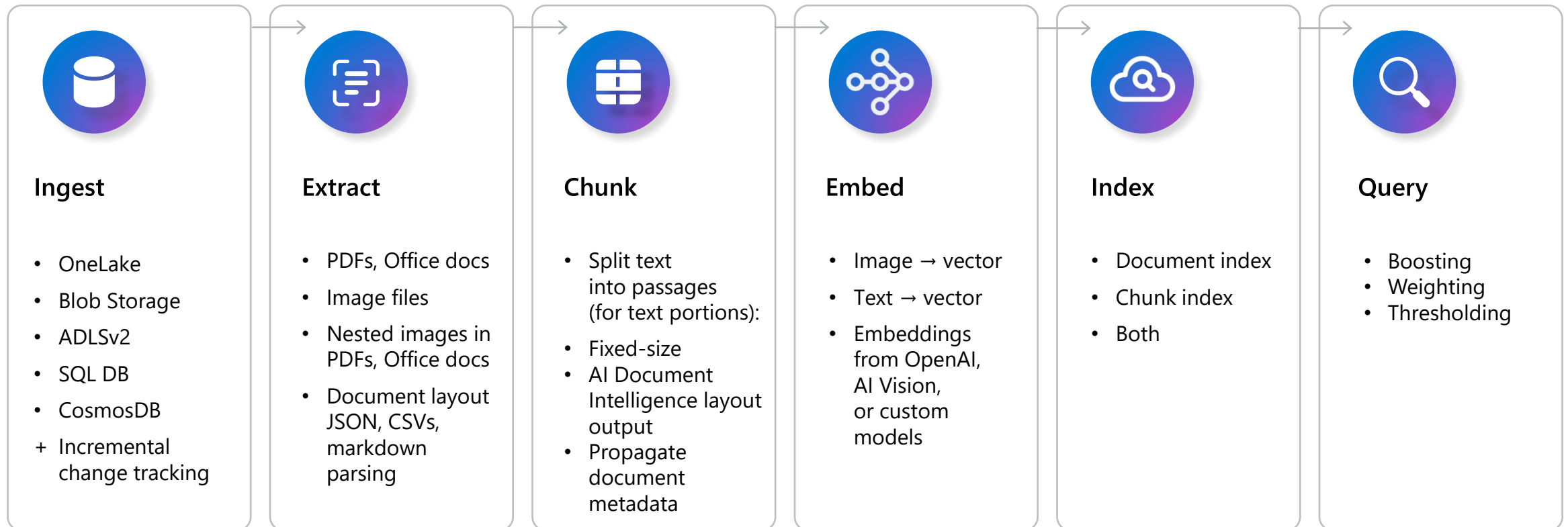
Anatomy of RAG



Data & Indexing

Streamlined RAG pipeline

Integrated vectorization and Azure AI Search



Data ingestion

Consider:

Data sources

Data preparation

Understanding
your data

Ingestion mechanisms
(Pull/Push)

Data AI enrichment

Chunking

Vectorization

Incremental indexing

Deletion policies

Scheduling

Security

Data and
service limits

Scalability

Cost

Error handling
and logging

Compliance

Data ingestion & connectors



Built-in pull indexer support for Azure data sources including OneLake, Blob Storage, Azure SQL, Cosmos DB



Data sources supported by Microsoft Partners: [BA Insights](#) has 92 connectors to pull data from including SharePoint Online, Confluence, OpenText



Push API/SDK for any data source not supported with pull method:

[Supported data sources](#)
[Integrated Vectorization](#)

[Data import and ingestion](#)
[Push SDK in RAG](#)

Streamlined RAG pipeline

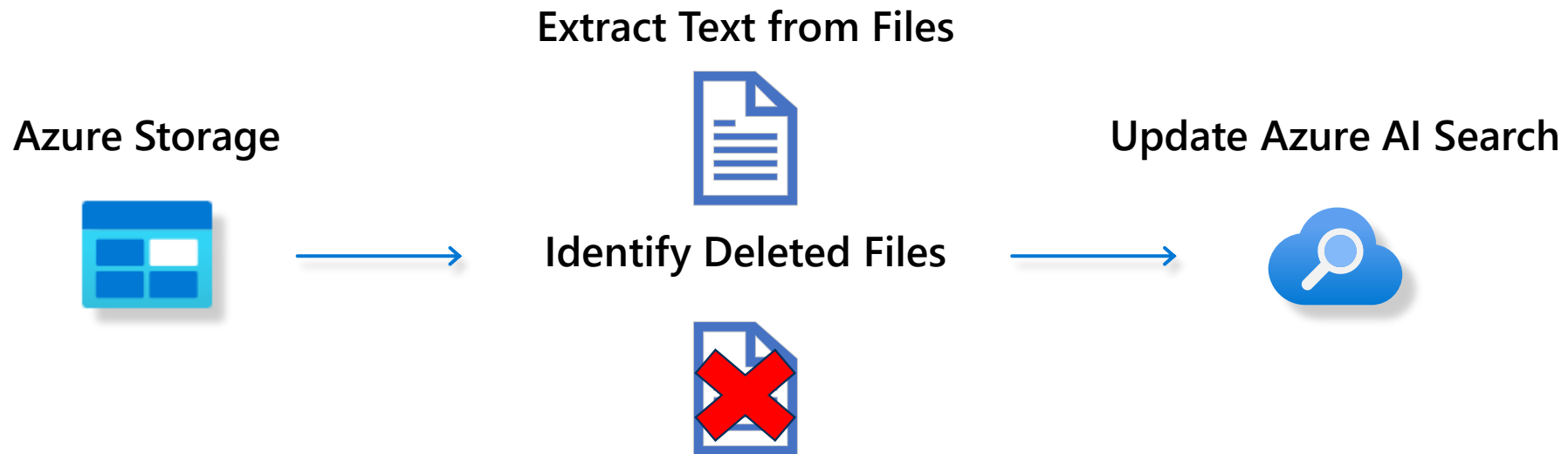
Supported data sources include

- Azure Storage
 - [Blob](#)
 - [Data Lake Storage Gen2](#)
 - [Table](#)
 - [Files](#)
- [OneLake](#)
- Azure Cosmos DB
 - [NoSQL](#)
 - [Gremlin](#)
 - [MongoDB](#)
- [Azure SQL](#)
- [Azure Database for MySQL](#)
- A variety [of partner-supported data sources](#)

Data sources (1/3)

Azure storage

- Automatically map supported files in [Azure Blob Storage](#), [OneLake](#) datalake, [Data Lake Storage Gen2](#), [Azure Files](#) to Azure AI Search
- Extract text from document formats such as EPUB and PDF
- Track changes and deletions to your files

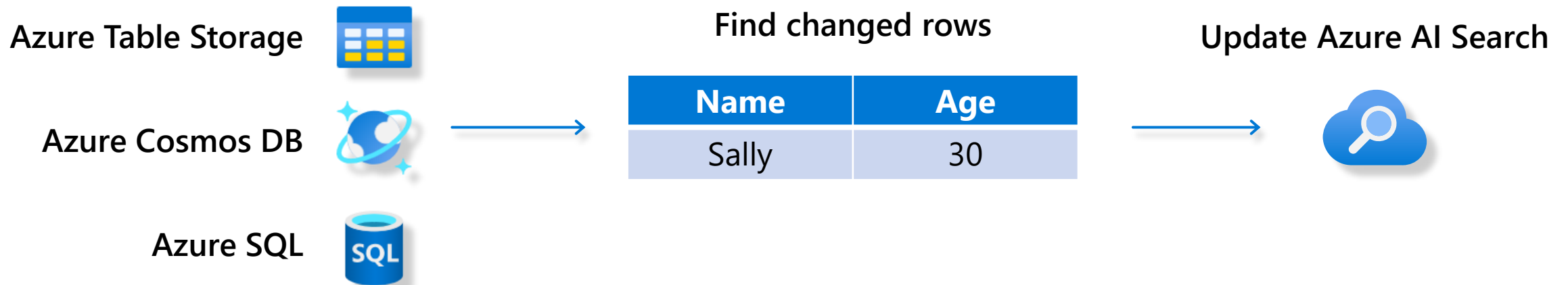


Data sources (2/3)

Other built-in data sources include

- Azure Table Storage
- Azure Cosmos DB for [NoSQL](#), [MongoDB](#), and [Apache Gremlin](#)
- [Azure SQL Database](#), [Azure SQL Managed Instance](#), or [SQL Server on Azure Virtual Machines](#)

Supports change and deletion tracking



Data sources (3/3)

Partner data sources

- Find over 100 data source connectors provided by third-party Microsoft partners

Chunking

Chunking

- Split long-form text into short passages
 - LLM context length limits
 - Focused subset of the content
 - Multiple independent passages
- Basics
 - ~200–500 tokens/passage
 - Maintain lexical boundaries
 - Introduce overlap
- Layout
 - Layout information is valuable, e.g., tables, content-structure preservation

Vectorization

- Indexing-time: convert passages to vectors

 [Example: Data preparation process \(including Chunking\)](#)

Your document Chunking strategy matters

Chunking solves 3 problems
for generative AI applications:

Allows multiple retrieved
documents

to be passed to the LLM within its
context window limit

Provides a mechanism for the most
relevant passages of a given
document

to be ranked first

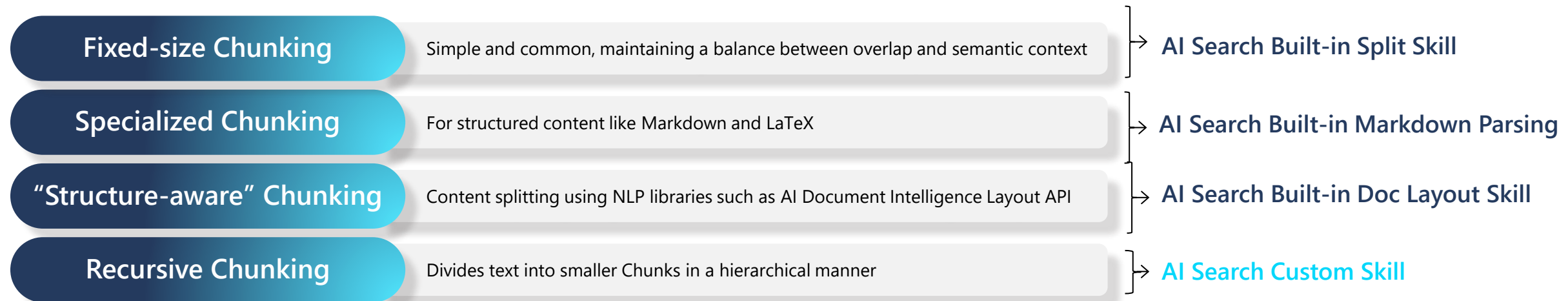
Vector search has a per-model limit
to how much content can be
embedded into each vector

Retrieval Configuration	Single vector per document <i>[Recall@50]</i>	Chunked documents <i>[Recall@50]</i>
Queries whose answer is in long documents	28.2	45.7
Queries whose answer is deep into a document	28.7	51.4

Chunk boundary strategy	Recall@50
512 tokens, break at token boundary	40.9
512 tokens, preserve sentence boundaries	42.4
512 tokens with 10% overlapping chunks	43.1
512 tokens with 25% overlapping chunks	43.9

Retrieval Configuration	Recall@50
512 input tokens per vector	42.4
1024 input tokens per vector	37.5
4096 input tokens per vector	36.4
8191 input tokens per vector	34.9

Chunking strategies



Preprocessing data
for quality

Selecting a range of potential
Chunk sizes considering
content nature and embedding
model capabilities

Evaluating performance
of each Chunk size

Indexing (Pull indexers)

Indexer (pull model)

- Crawler that extracts data from cloud data sources and populates a search index
- Drive skillset execution (AI enrichment)
- Can be run on demand or on a recurring data refresh schedule (incremental indexing)

Best practices

- Understand indexing limits
- Run on a schedule and continuously to correct errors
- Identify data source needs (multiple data sources going to a single index, data source support, limits and configuration)
- Configure data source document deletion policy if needed
- Recognize data types for proper parsing (JSON, JSON lines, CSV, markdown, etc.)
- If handling PDFs or images with embedded text, consider AI enrichment with OCR or Azure AI Document Intelligence layout
- Configure batch size based on workload
- Consider integrated Vectorization if using vector search
- Identify AI enrichment needs, such as language translation or recognition, key phrase extraction, etc.
- Test with various data types/sizes to understand indexing time and optimize the process
- Identify type of authentication being used (Managed identity is recommended)
- Recognize if private endpoints will be used for secure connections and configure shared private links appropriately

 [Indexing example with Integrated Vectorization](#)

Indexing (Push method)

- Push method uses APIs to upload documents into a search index
- No restrictions on data source type or frequency of execution
- Connectivity and the secure retrieval of documents under your control

Key benefits

Versatility: JSON documents from any source can be uploaded. You can use AI Document Intelligence layout model, an LLM like GPT-4o for data parsing or your preferred parsing library

Flexibility: No restrictions on frequency of data push (no specific schedule, just API restrictions apply)

Control: Retrieval of documents under your own control

Pushing data to Azure AI Search index

- APIs for single or multiple documents upload
- Follow best practices for indexing large documents

Indexing actions

Upload: Insert new documents or update existing ones

Merge: Updates existing document, replacing values

Merge or upload: Behaves like merge for existing documents and like upload for new ones

Delete: Removes entire document from index

[!\[\]\(626ce8ac21792b9405bfddfea8e0c96a_img.jpg\) Push method sample for RAG](#)

Retrieval strategies



Keyword search

- **For exact, plain text matches**
- “Vocabulary gap” in Q&A systems like Copilot



Vector search

- **For conceptual similarity, or underlying meaning**
- Weak performance on exact matches (like a product ID or code)



Hybrid search

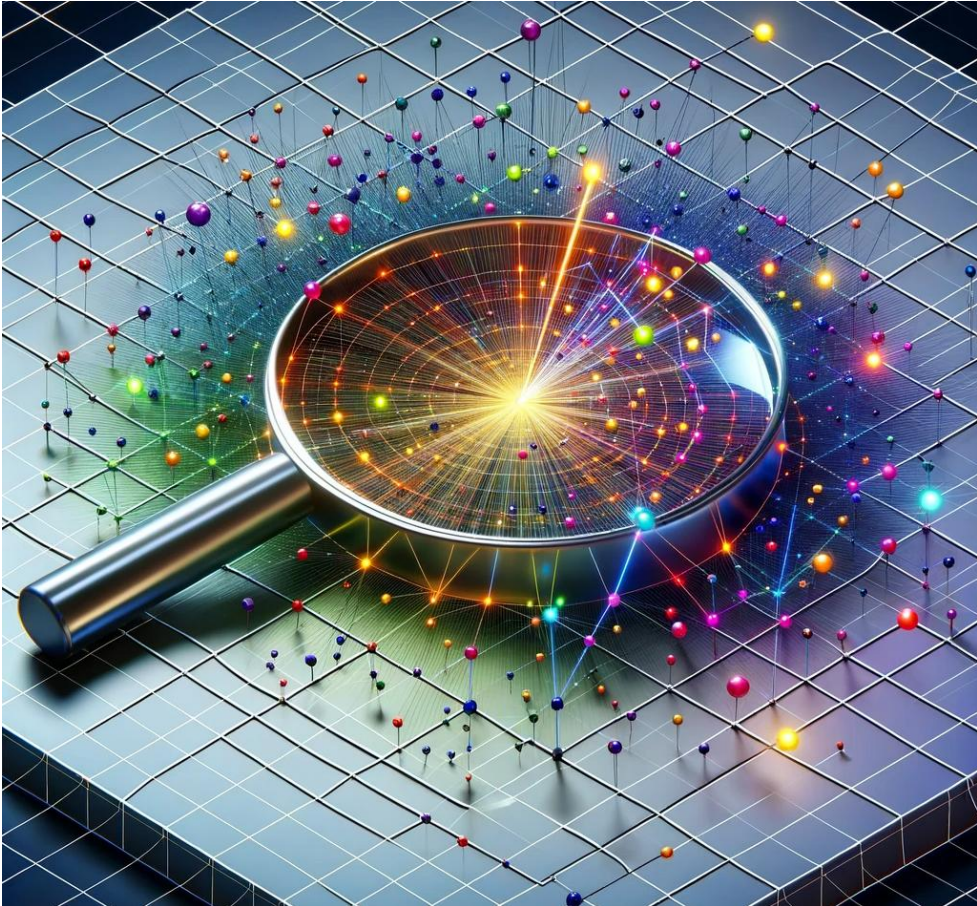
- **Best of both vectors and keywords**
- Brings more accurate responses across various scenarios



Search re-ranking

- **Scores and ranks all retrieved documents by relevance**
- Reranking runs after performing search strategy (can't retrieve information)

Vectors



Learned vector representations

- Models that encode item $\rightarrow$ vector
- Similar items map to close vectors
- Sentences, images, graphs, etc.

Vector search

- Find K closest vectors given a "query" vector
- Search exhaustively or through approximations

Vector search

- Full-featured
- Create embeddings using any model
- Explicit and transparent vector data processing
- Exhaustive KNN search & ANN search
- Multi-vector



Vector search strategies

- ANN search
 - Fast vector search at scale
 - Uses HNSW, a graph method with excellent performance-recall profile
 - Fine control over index parameters
- Exhaustive KNN search
 - Per-query or built into schema
 - Useful to create recall baselines
 - Scenarios with highly selective filters
 - e.g., dense multi-tenant apps

```
r = search_client.search(  
    None,  
    top=5,  
    vector_queries=[RawVectorQuery(  
        vector=search_vector,  
        k=5,  
        fields="embedding")])
```

```
r = search_client.search(  
    None,  
    top=5,  
    vector_queries=[RawVectorQuery(  
        vector=search_vector,  
        k=5,  
        fields="embedding",  
        exhaustive=True)])
```


Rich vector search query capabilities

Filtered vector search

- Scope to date ranges, categories, geographic distances, etc.
- Rich filter expressions
- Pre-/post-filtering
 - Pre-filter: great for selective filters, no recall disruption
 - Post-filter: better for low-selectivity filters, but watch for empty results

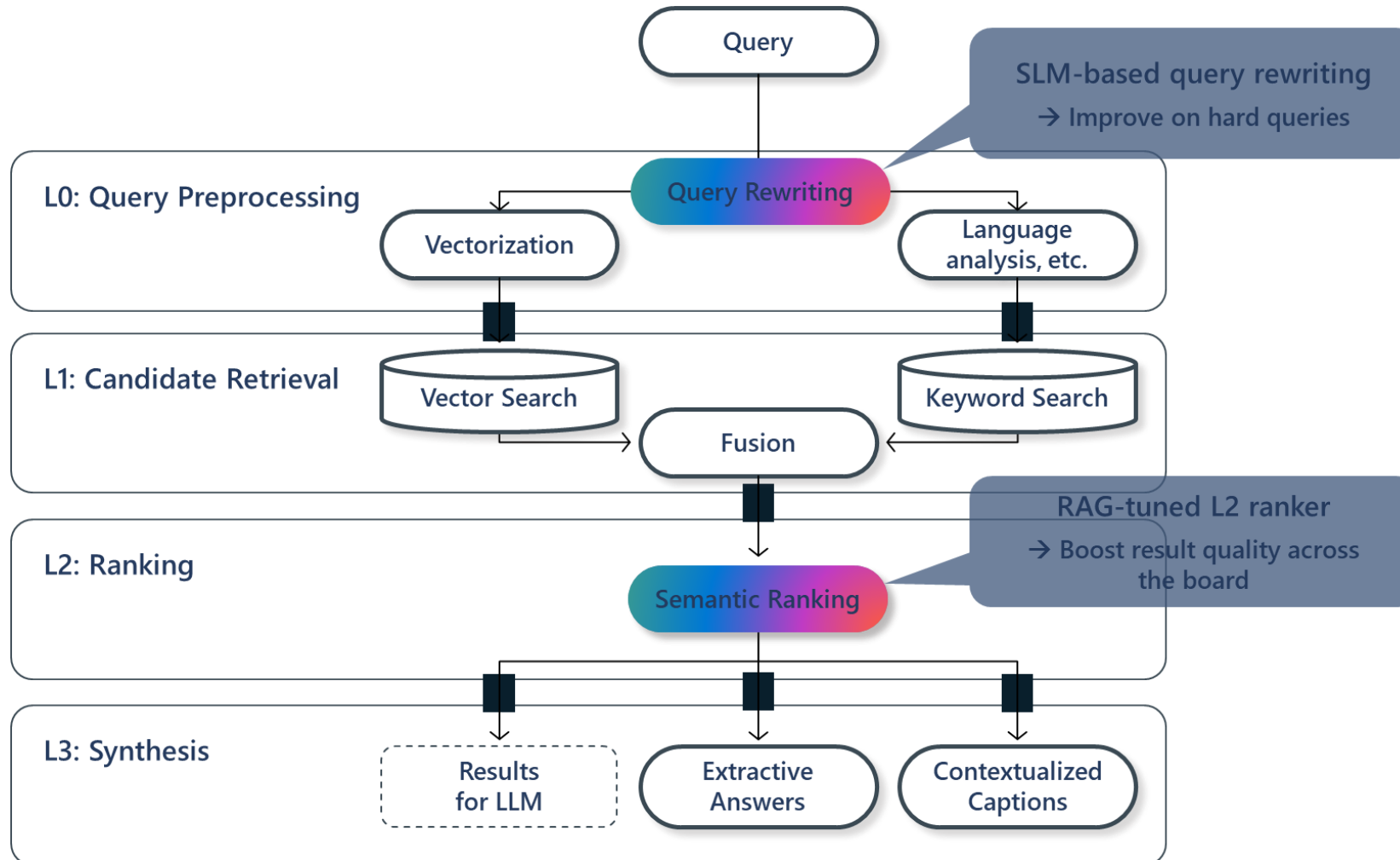
Multi-vector scenarios

- Multiple vector fields per document
- Multi-vector queries
- Can mix and match as needed

```
r = search_client.search(
    None,
    top=5,
    vector_queries=[RawVectorQuery(
        vector=query_vector,
        k=5,
        fields="embedding")],
    vector_filter_mode=VectorFilterMode.PRE_FILTER,
    filter=
    "category eq 'perks' and created gt 2023-11-15T00:00:00Z")
```

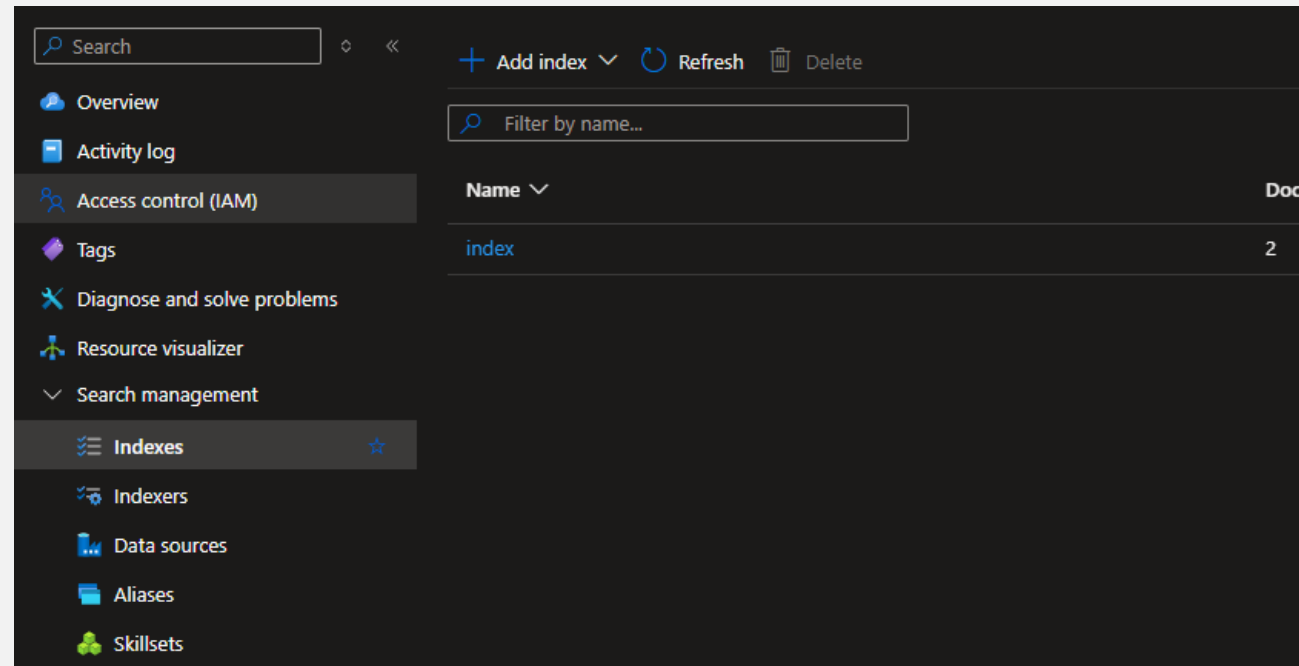
```
r = search_client.search(
    None,
    top=5,
    vector_queries=[
        RawVectorQuery(
            vector=query1, k=5, fields="embedding"),
        RawVectorQuery(
            vector=query2, k=5, fields="embedding")
    ])
```

Query Engine in practice



Demonstration

Demo the Search Service through the Azure Portal



OPTIONAL Lab - RAG

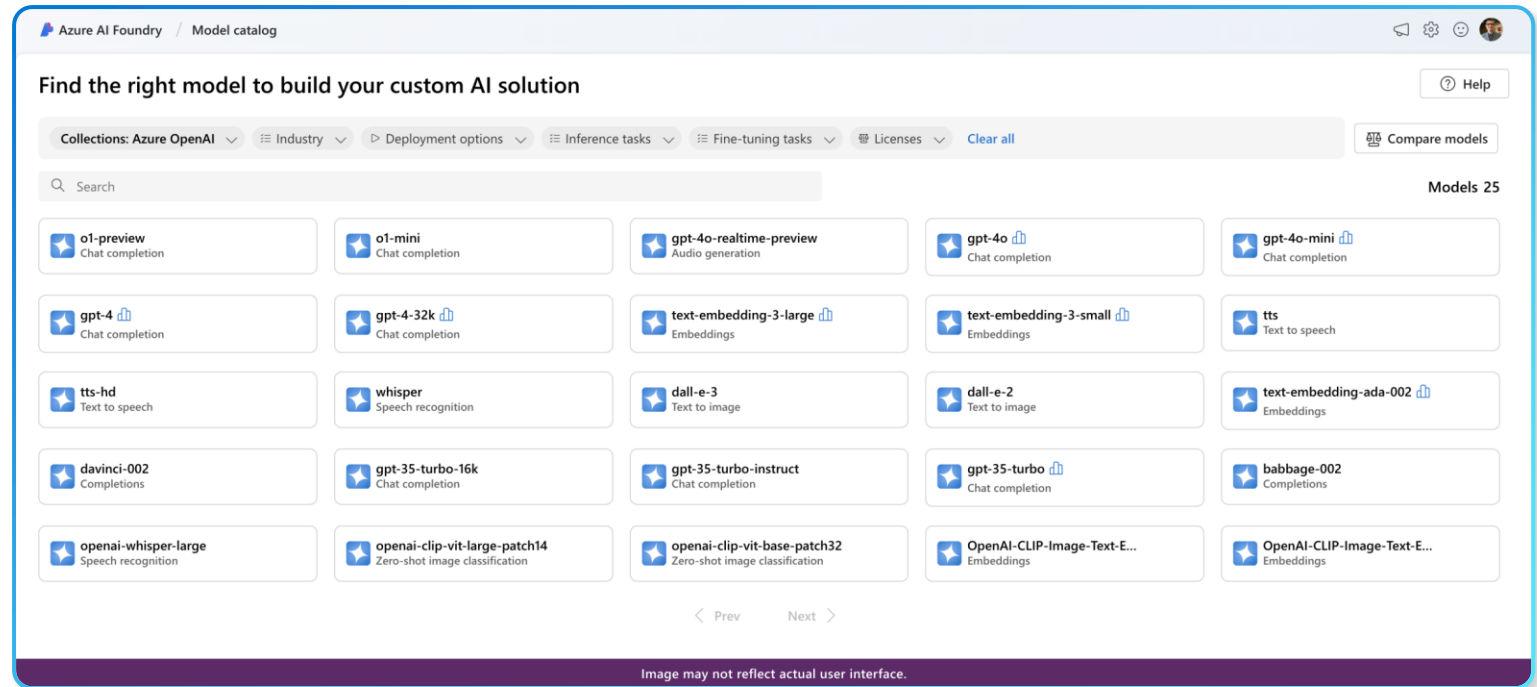
Lab Description

- In this lab, we will implement a RAG (Retrieval-Augmented Generation) system using Azure AI Search and Azure OpenAI. RAG is a technique that combines information retrieval with text generation, allowing language models to access specific external knowledge beyond their training.
- We will explore:
 - **Query without RAG:** How the LLM responds without additional knowledge
 - **Azure AI Search Setup:** Preparing the vector search index
 - **RAG Implementation:** Integrating search with response generation
 - **Results Comparison:** Demonstrating the difference between responses with and without RAG

Azure OpenAI

Azure OpenAI Service (AOAI)

- Widest Range of Deployment offerings - Standard, Provisioned and Batch available across 28 different regions, data zones and Global
- Your data is private and secure. Data residency at rest and at processing, and data protection from exfiltration.
- Safety is built-in with tools to responsibly build and implement AI.
- Enterprise Azure promises with private networking, managed identity, customer managed keys, azure monitoring, cost management, and more.
- Developer-Friendly Integration. Seamless integration with GitHub, Visual Studio and Copilot Studio.
- Choice of Models - Over 11,000 models available to integrate into your application.



Leverage OpenAI's state-of-the-art models and Azure's cloud platform to build innovative AI-driven applications and solutions across various industries and domains.

Azure OpenAI Service (AOAI)



Cutting-edge AI models
Pioneering research and advancements
Unlocking new creative possibilities



Azure AI Foundry

Secure data and built-in responsible AI
Global availability and flexibility
Developer-friendly integrations

Azure OpenAI Service

Connect your data | Customize your agents | Manage quota & deploy

Reasoning and
multimodal
text, image, audio

o1
o1-mini
GPT-4o
GPT-4o mini

Fine-tuning

GPT-4o
GPT-4o mini

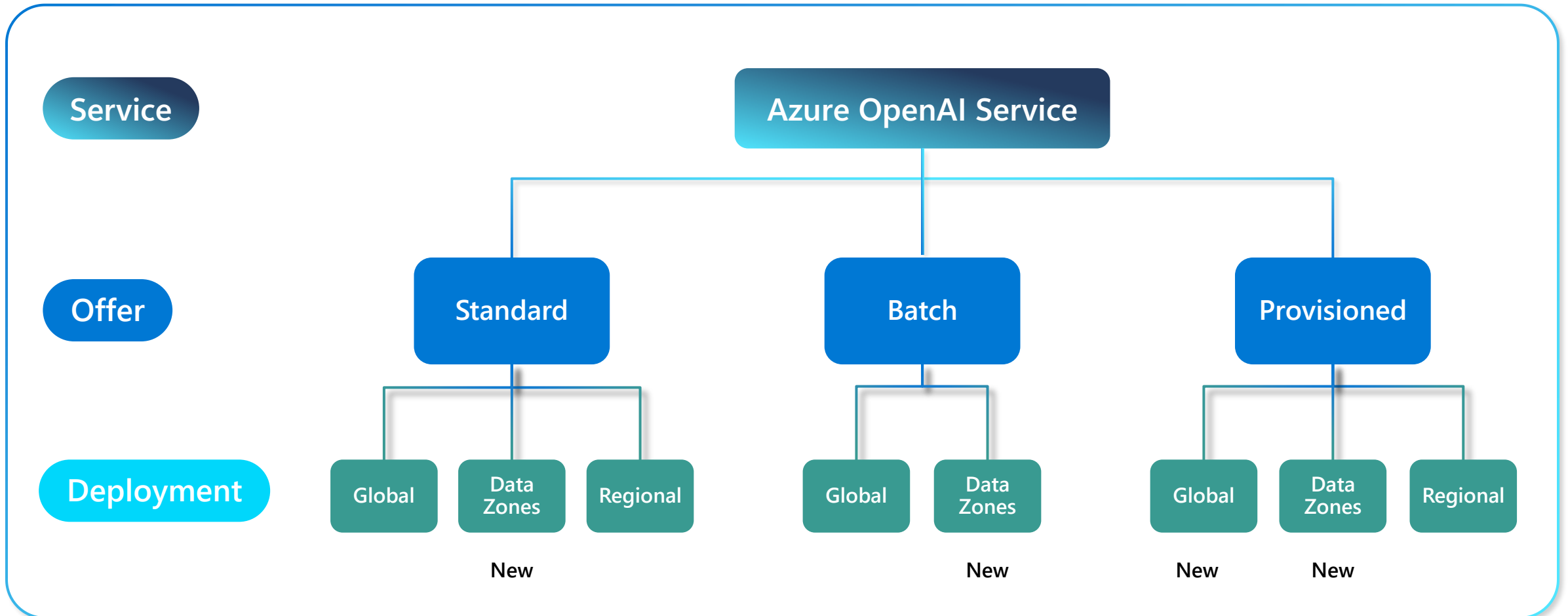
Images

DALL·E 3

Transcription &
translation

Whisper

Azure OpenAI Service Offerings



Azure OpenAI Service - GPT-4o-Realtime-Preview

Enable low-latency, multilingual speech-to-speech conversations

Real-time Audio
Interactions

Natural Speech
Generation

Multilingual
Capabilities

Azure OpenAI Service - GPT-4o-Audio-Preview

Enable advanced audio processing and content generation

Native sentiment
analysis

Immersive Speech
Generation

Multilingual
Capabilities

Azure OpenAI Service – o1 Models

o1

Advanced reasoning

Solves complex problems

Enhances LLM capabilities:

- Logical problem-solving
- Complex math
- Scientific inquiries
- Extended conversations

o1-mini (in preview)

Smaller, faster model

Excels at:

- Instruction following
- Small context operations
- Code debugging

80% cheaper than o1-preview

Text Based
Input/Output Only

Generates 6-10x
more tokens

Support for 128K
Context window

Longer response
time for reasoning

Azure OpenAI Service model guidance

o1

Multimodal model focused on advanced reasoning and solving complex problems, including math and science tasks.

o1-mini

Smaller, faster, and more cost-efficient version of o1. Performs well at instruction following and small context operations.

GPT-4o

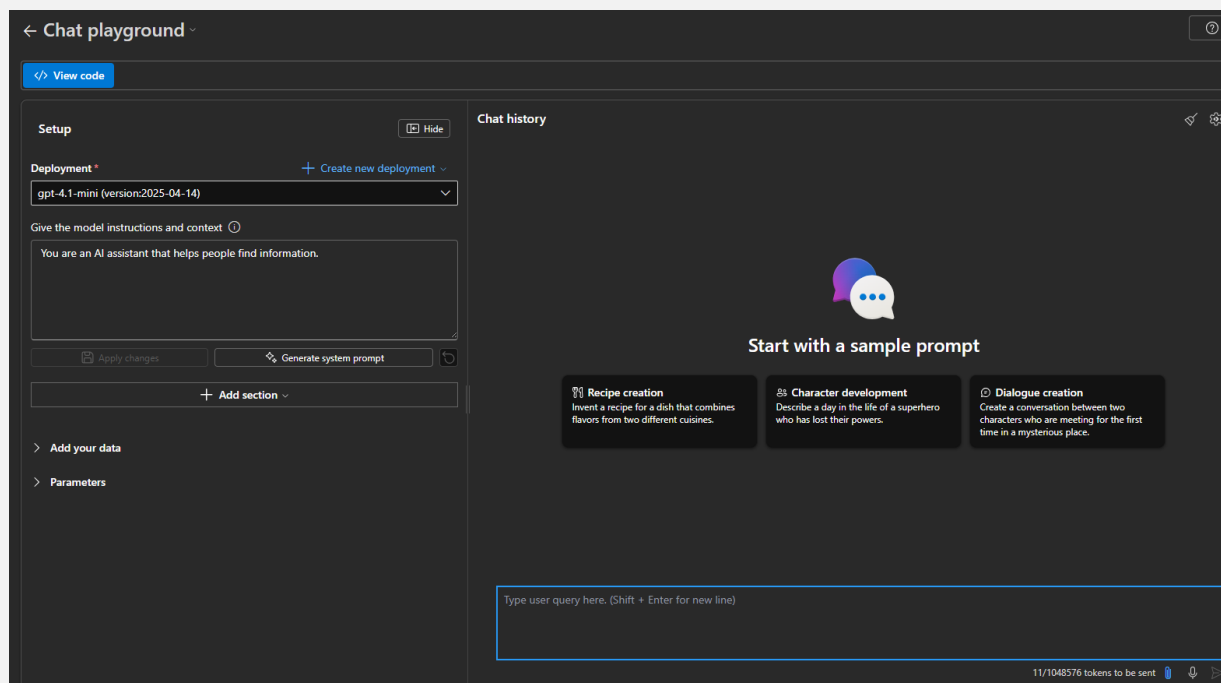
A versatile, multimodal model for text and image processing, non-English languages, and vision tasks.

GPT-4o-mini

A cost-effective version of GPT-4o for lightweight applications.

Demonstration

Demo the Azure OpenAI Service through the Azure AI Foundry Studio



Azure AI Content Safety

Generative AI introduces new risks



Ungrounded
outputs & errors



Jailbreaks &
prompt injection
attacks



Harmful content
& code



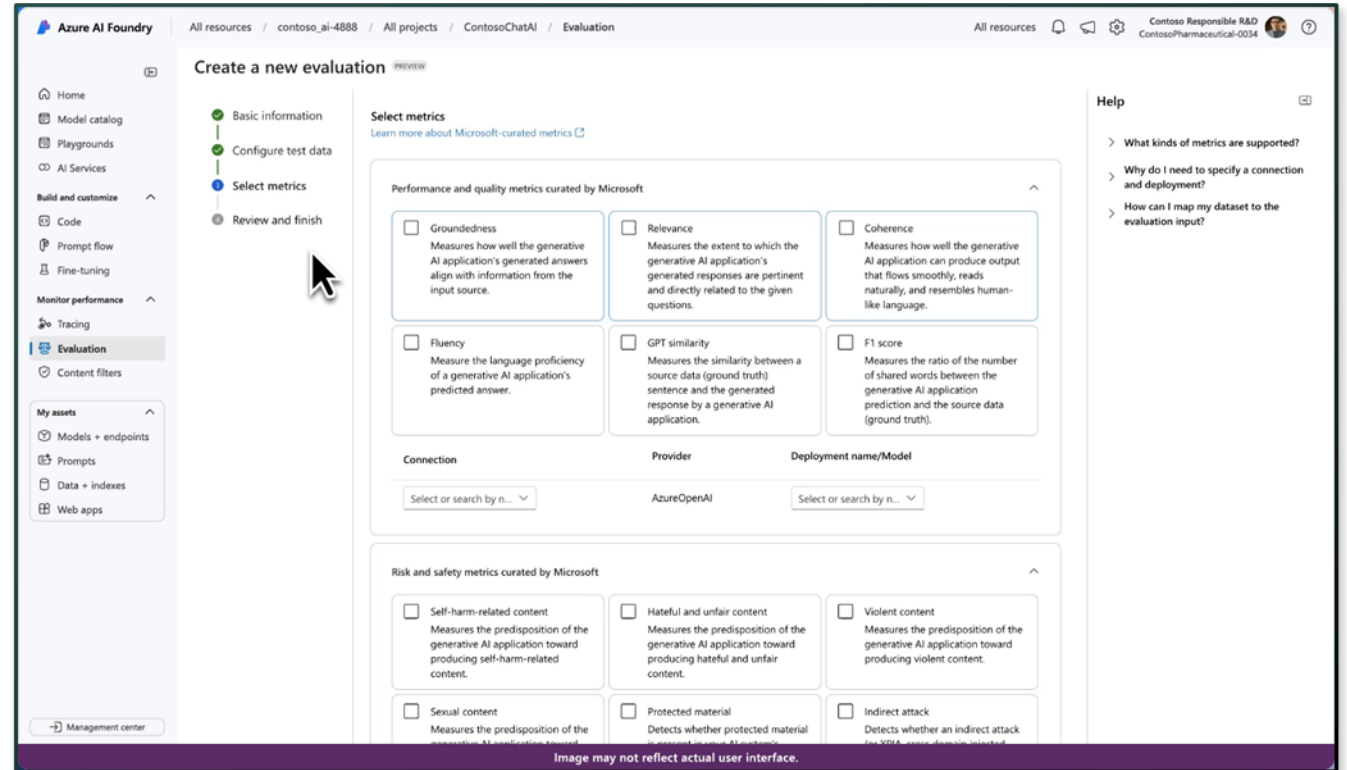
Copyright
infringement



Manipulation
& human-like
behavior

Safeguard with Trustworthy AI

- Secure your data, AI apps, and models
- Build end-to-end verifiable privacy into your AI apps
- Utilize built-in AI safety tooling



Safeguard with Trustworthy AI

Use and build AI that is secure, private, and safe.

Secure your data, AI, apps, and models

Confidently design apps with technologies, templates, and governance best practices to help manage security risk and compliance.

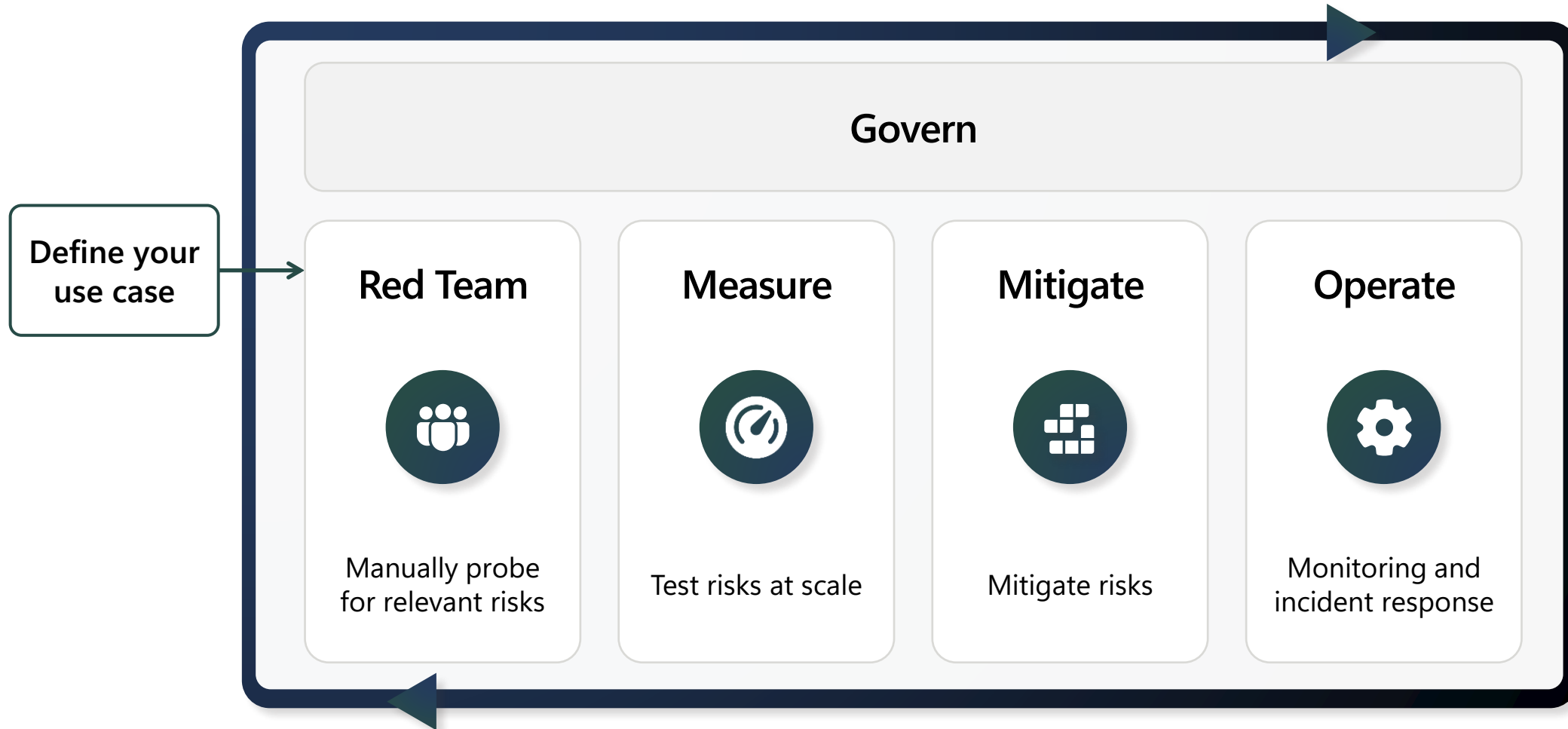
Build end-to-end verifiable privacy into your AI apps

Protect your data at rest, in transit, and in use with confidential AI.

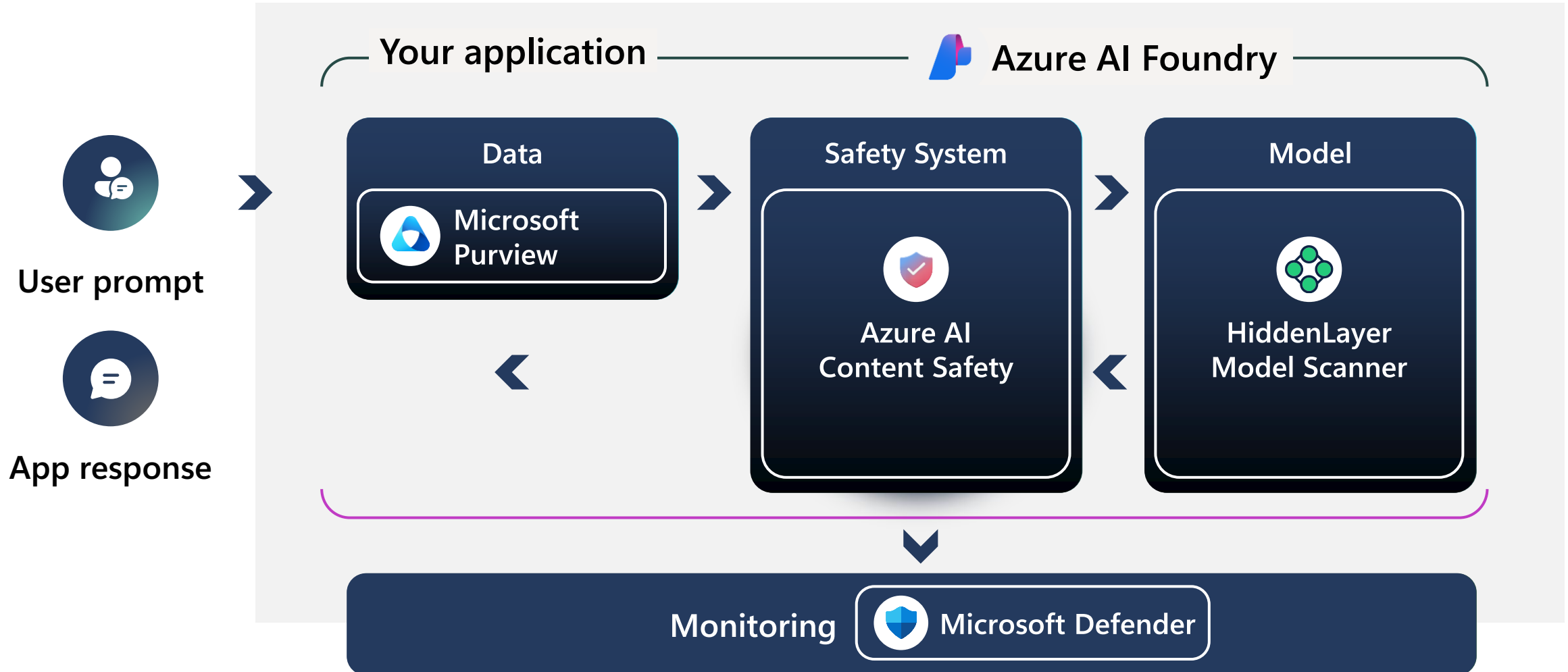
Utilize built-in AI safety tooling

Detect and filter harmful human and AI-generated content, protect PII, and safeguard applications against prompt attacks, hallucinations, protected and owned materials, and bias.

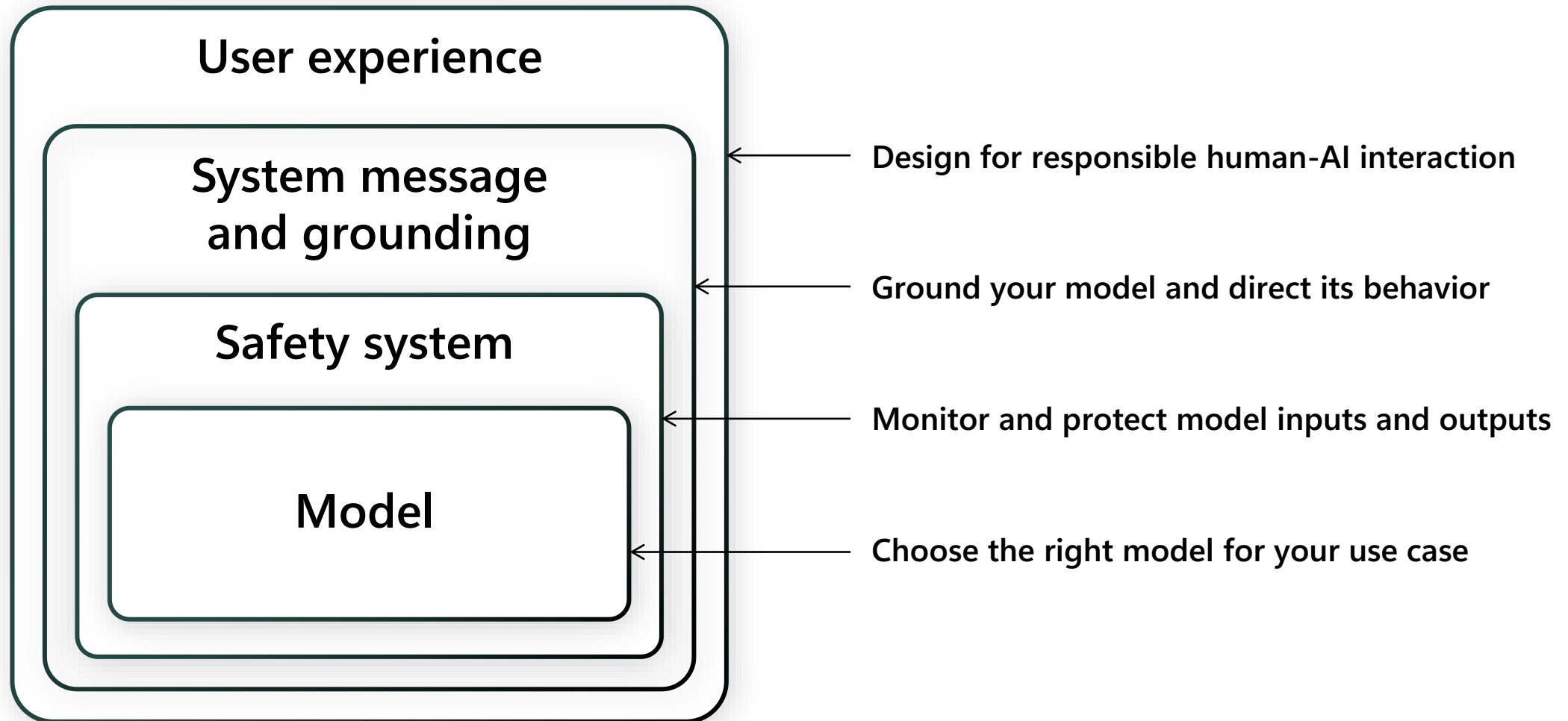
Responsible innovation is iterative



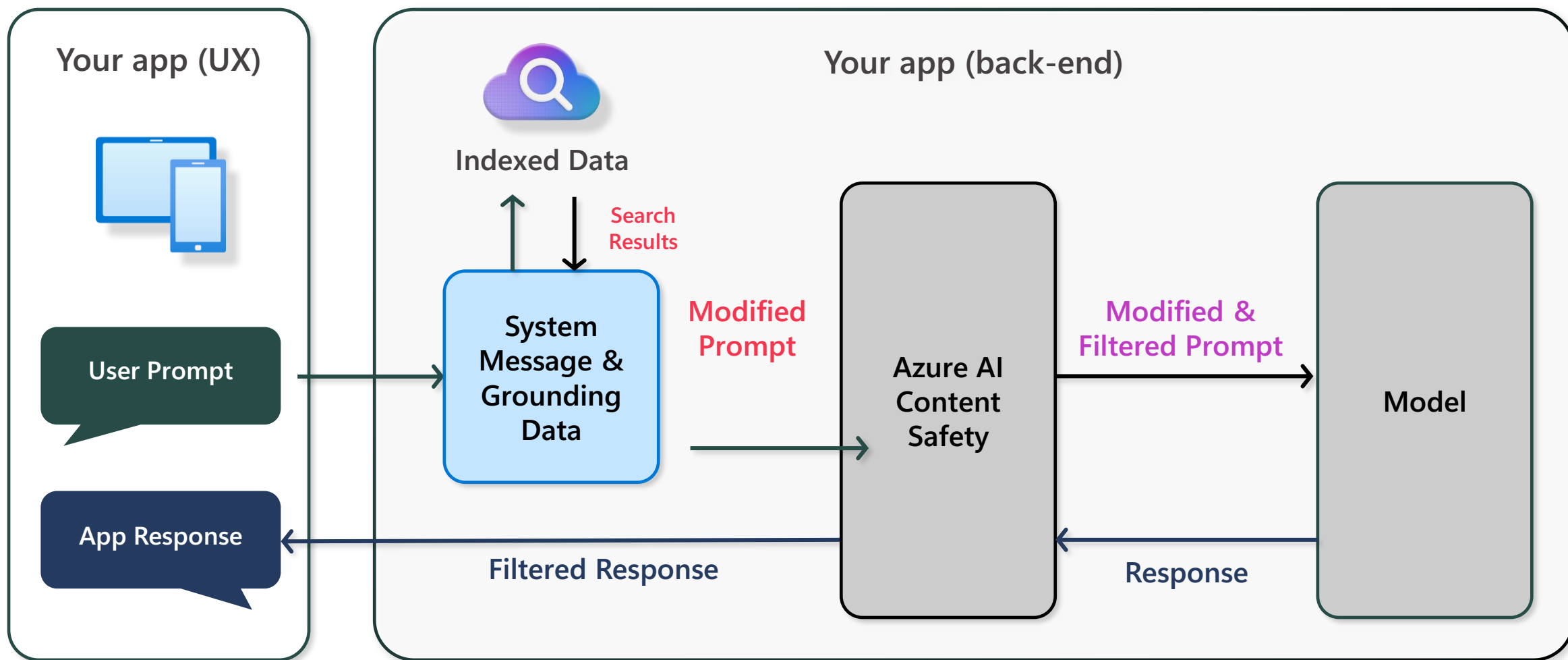
Built-in security and safety system



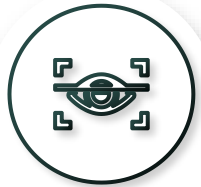
Risk mitigation layers



How these mitigations happen in real time

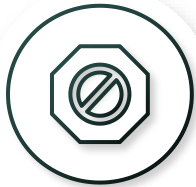


Responsible tooling for confidence



Multimodal filtering

Scan text, images and multi-media to identify, block and monitor **harmful content**



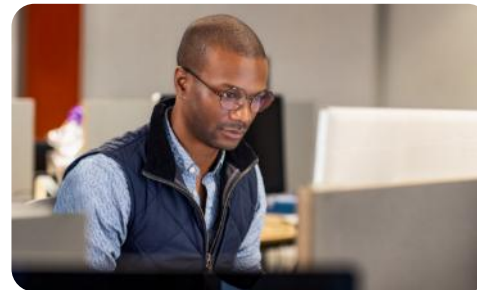
Customized systems

Create blocklists and **custom categories** to block entire topics, not just specific words



Prompt shielding

Identify and mitigate prompts that could expose you to **prompt injection attacks**



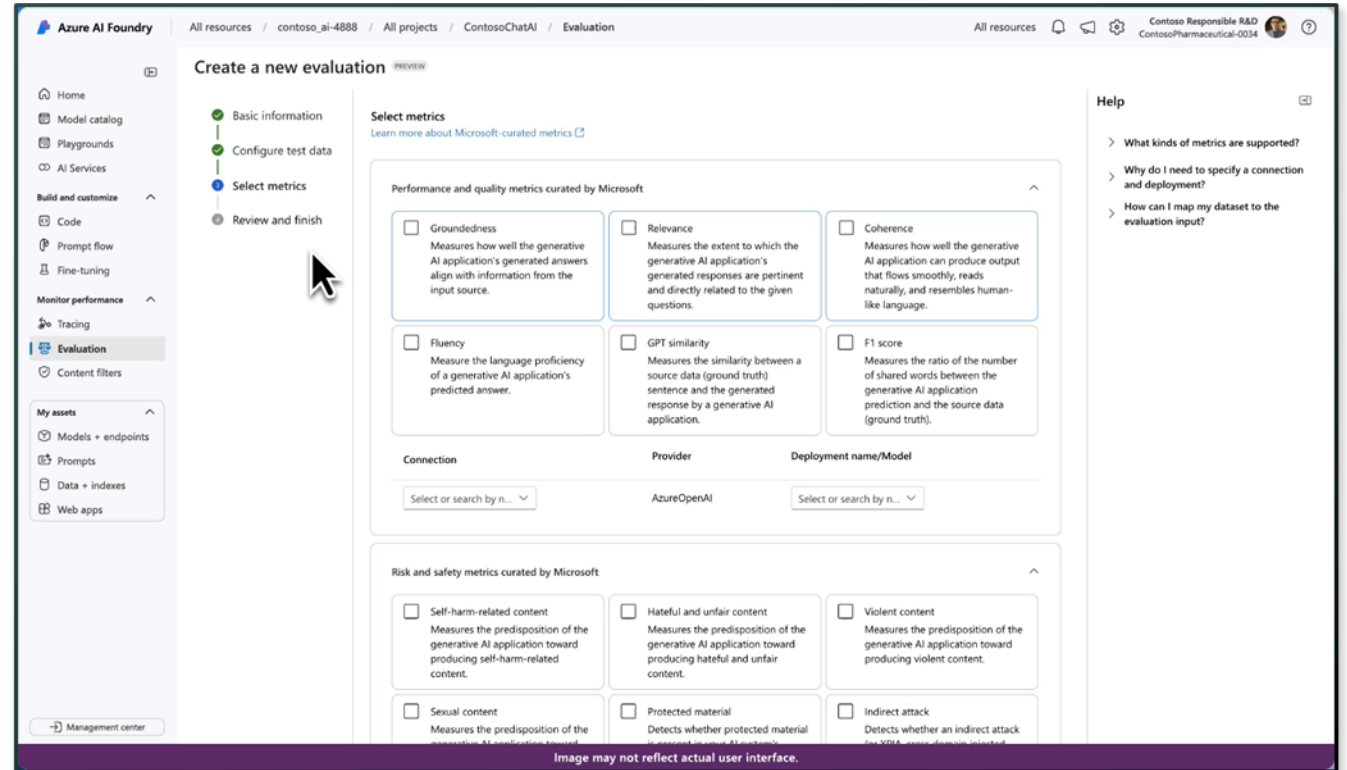
Protected materials

Avoid outputting **known or owned text** content with protected materials detection



Safeguard with Trustworthy AI

- Secure your data, AI apps, and models
- Build end-to-end verifiable privacy into your AI apps
- Utilize built-in AI safety tooling

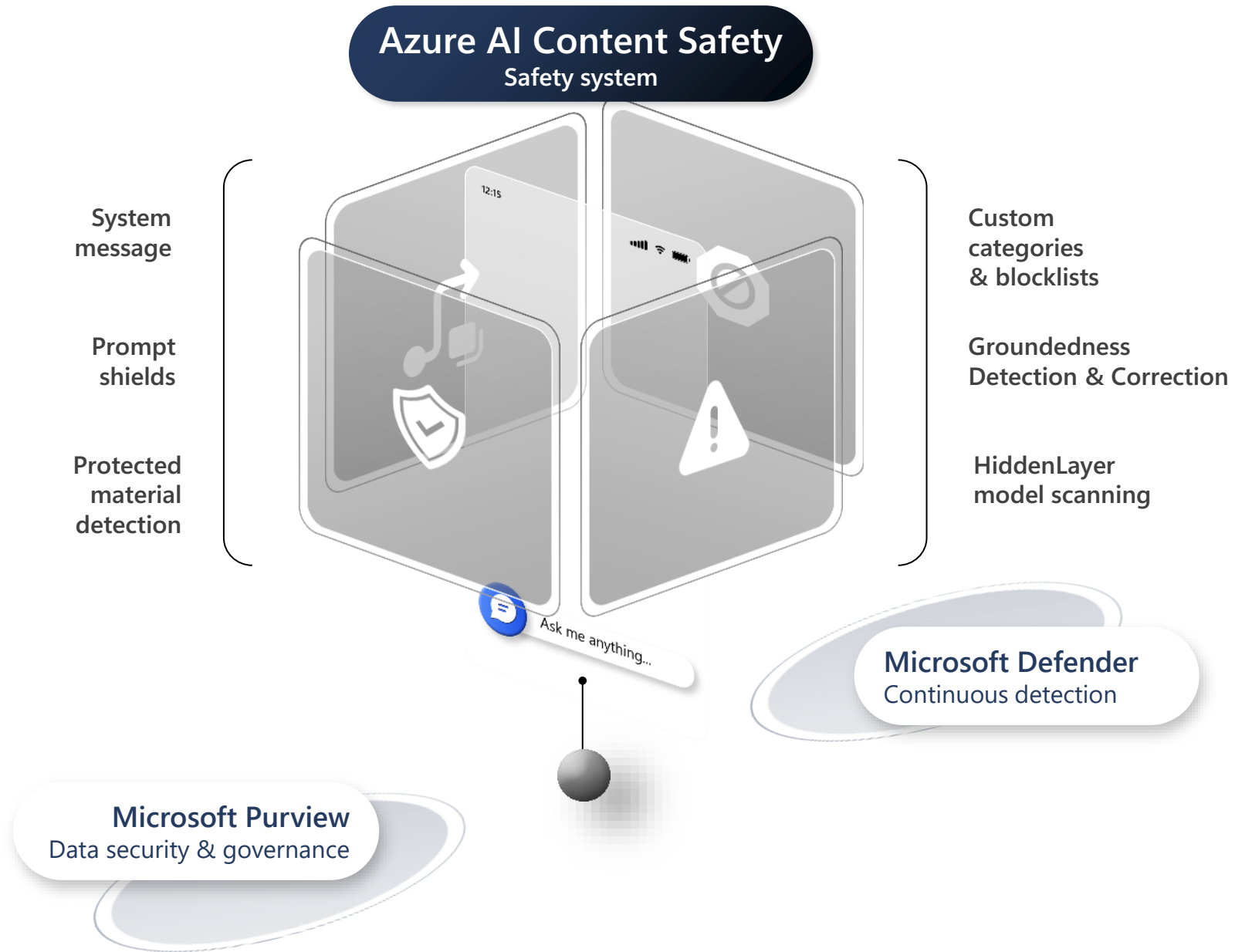


Azure AI Content Safety

Drive innovation
with a built-in, comprehensive
safety system

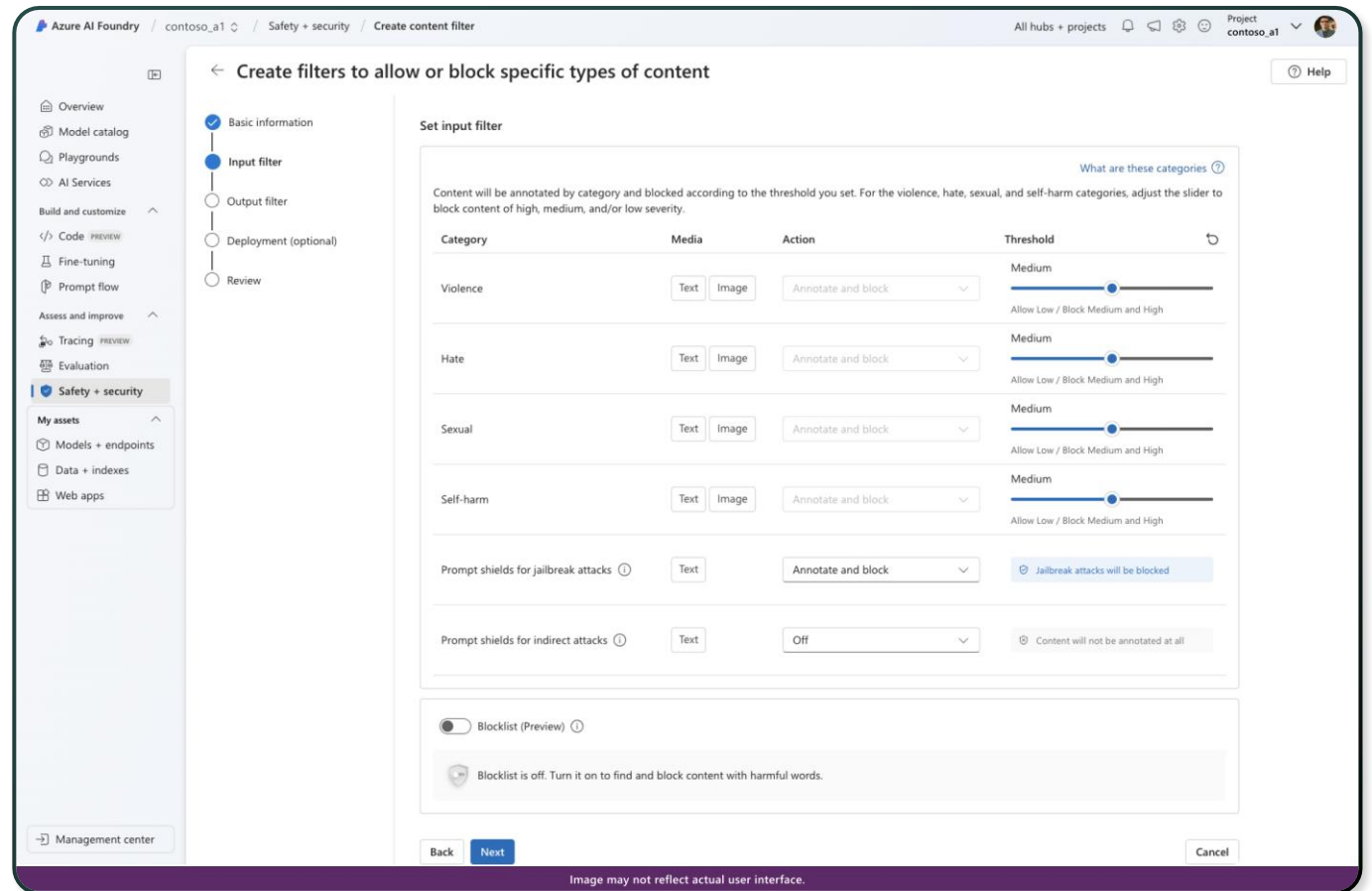
The Azure AI Foundry difference:

- Security and governance
- HiddenLayer model scanning
- Copyright commitment



Azure AI Content Safety

Azure AI Content Safety is a content moderation platform that uses AI to keep your content safe, creating better online experiences for everyone with powerful models that detect inappropriate content.



Enhance user safety and data security with Azure AI Content Safety, providing tailored protection, real-time threat detection, and customizable controls.

Responsible by Design

Because AI principles are not self-executing, we share our learnings and embed data-driven guardrails, guidance and best practices into Azure AI Foundry to help you operationalize trustworthy AI.

Microsoft's Responsible AI principles



Fairness



Reliability
& Safety



Privacy
& Security



Inclusiveness



Transparency



Accountability

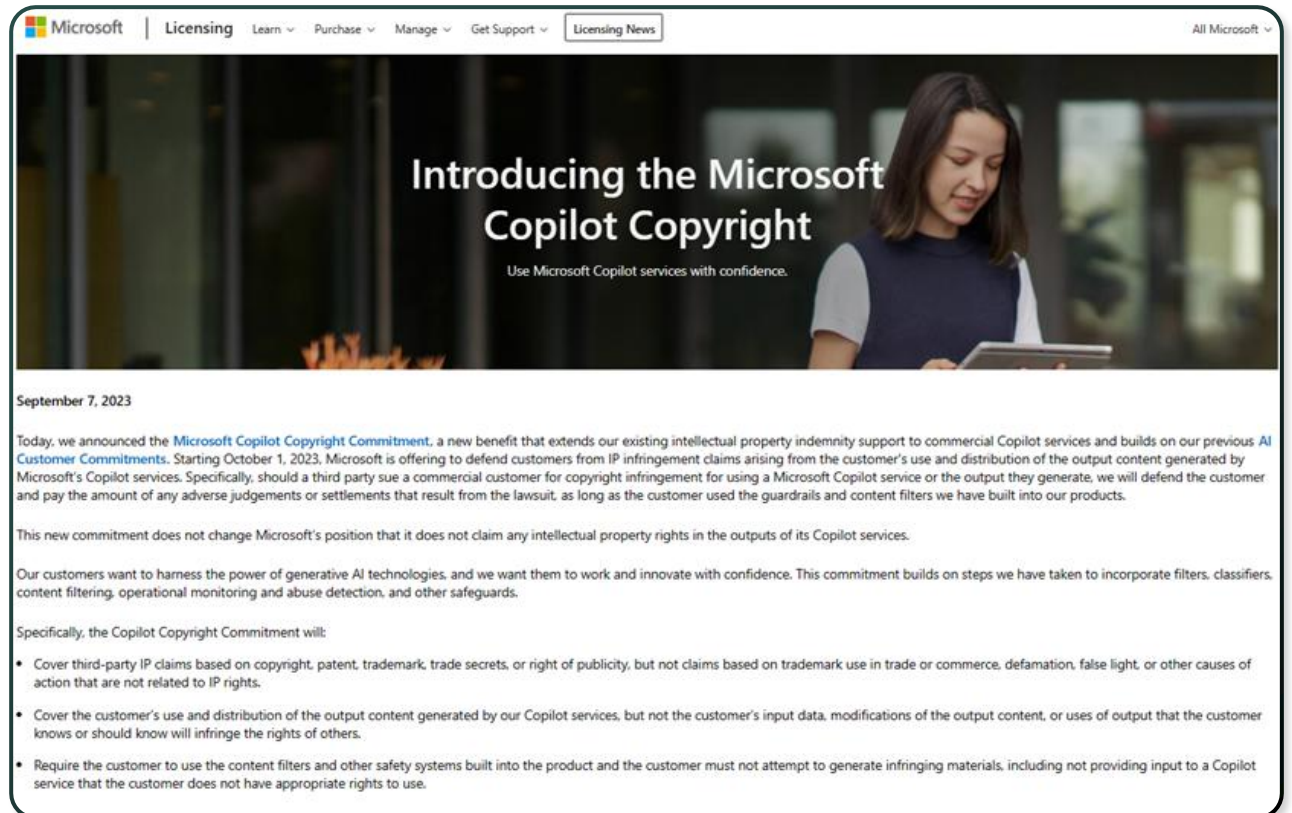


Microsoft first adopted our six AI principles in 2018, and they continue to drive our policy, research, and engineering investments.

Microsoft Copilot Copyright Customer Commitment

Copilot Copyright Customer Commitment ensures defense against IP infringement claims arising from Copilot services to enhance customer confidence.

- Microsoft defends customers against copyright claims, settling adverse judgements and settlements
- Microsoft protects output content generated by Copilot services, requiring adherence to built-in guardrails and content filters
- Microsoft address copyright, patent, trademark, and trade secrets claims, excluding claims unrelated to IP rights
- Microsoft requires use of content filters and other safety systems built into the product
- Microsoft helps facilitate innovation by defending against IP claims, fostering customer confidence in utilizing Copilot services for creative endeavors



The screenshot shows the Microsoft Licensing page with a navigation bar including links for Licensing, Learn, Purchase, Manage, Get Support, and Licensing News. The main headline reads "Introducing the Microsoft Copilot Copyright" with the subtext "Use Microsoft Copilot services with confidence." Below the headline is a date "September 7, 2023" and a paragraph announcing the "Microsoft Copilot Copyright Commitment" as a new benefit extending intellectual property indemnity support to commercial Copilot services. It states that starting October 1, 2023, Microsoft will defend customers from IP infringement claims arising from the customer's use and distribution of output content generated by Microsoft's Copilot services. The announcement specifies that Microsoft will defend the customer and pay the amount of any adverse judgements or settlements that result from the lawsuit, as long as the customer used the guardrails and content filters built into the products. It also notes that this commitment does not change Microsoft's position that it does not claim any intellectual property rights in the outputs of its Copilot services. The page further states that customers want to harness the power of generative AI technologies and work with confidence, and that this commitment builds on steps taken to incorporate filters, classifiers, content filtering, operational monitoring and abuse detection, and other safeguards. Specifically, the Copilot Copyright Commitment will:

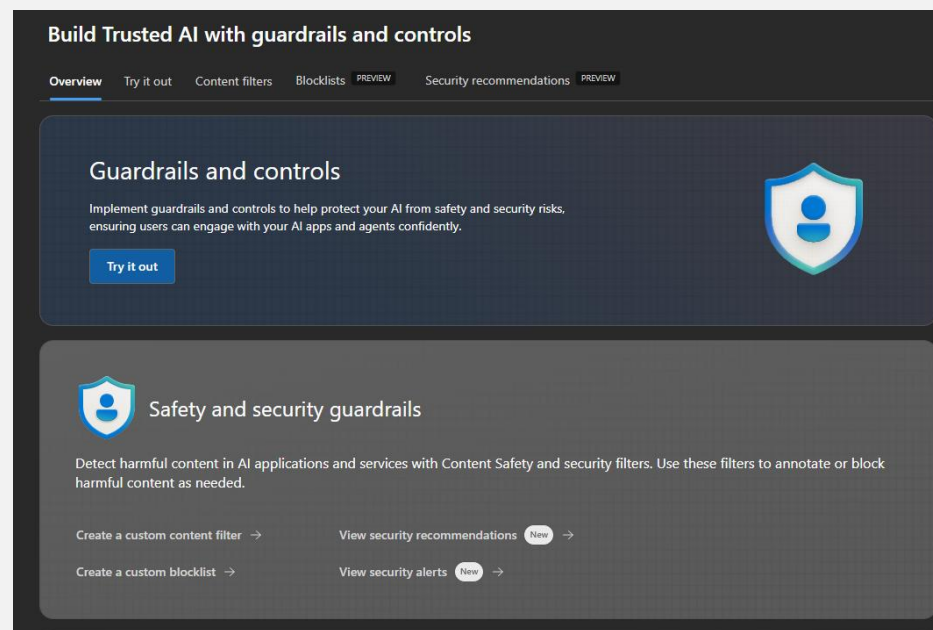
- Cover third-party IP claims based on copyright, patent, trademark, trade secrets, or right of publicity, but not claims based on trademark use in trade or commerce, defamation, false light, or other causes of action that are not related to IP rights.
- Cover the customer's use and distribution of the output content generated by our Copilot services, but not the customer's input data, modifications of the output content, or uses of output that the customer knows or should know will infringe the rights of others.
- Require the customer to use the content filters and other safety systems built into the product and the customer must not attempt to generate infringing materials, including not providing input to a Copilot service that the customer does not have appropriate rights to use.



Microsoft empowers businesses to leverage Copilot services confidently, mitigating legal risks and fostering innovation within established boundaries.

Demonstration

Demo the Azure AI Content Safety tool through the Azure AI Foundry Studio



<https://ai.azure.com/resource/contentfilters/tryitout?>

Questions?

Thank You!